Finding What Changed

A difference-imaging and transient discovery pipeline for SPHEREx Data Release 1

SPHERExLens • 30-slide specification, abundant technical figures, and complete script
full route 28:10 • 15-minute route 18:40 • questions after

Evan Lim • November 2026

Deck map and production rules

30 slides on 32 pages. This section is the specification; pages 2–31 are the slides, one page each: the slide's own bullets, its figure brief, the figure itself, then the word-for-word script. Page 32 is a backup page with five technical questions and a pre-flight presentation checklist.

The full route runs **28:10**. The slides marked ✓ in the last column form a **18:40** route: fourteen slides dropped, core scientific argument intact. **The live demonstration is slide 29** (1:30) and uses the browser app (`site/index.html` or the deployed Netlify/Cloud Run app) already open in a second tab at 100% zoom, with Barnard's Star loaded – plus Live IRSA mode if showing on-demand fetching.

Every code metric, completeness rate, and detection count in this deck is verified directly against the current runtime modules (modular packages in `spherexlens/`, `step1`–`step7`, support modules, `app.py`, `static/`) and the live Cloud Run snapshot API (`/api/targets`, `/api/dashboard`, `/api/metrics`, `/api/deep`, `/api/snapshot-history`).

Page	Slide	Title	Dominant visual	Time	15 min
2	1	Title and the headline results	4-panel epoch difference, hero numbers	0:45	✓
3	2	Why difference imaging: the physics of change	4 panels: visit 1, visit 2, difference with dipole/fading, and flaring monopole	0:55	✓
4	3	Problem formulation and operational constraints	Dataflow from 32,400 pixels to threshold to candidate set and 4 constraint cards	1:00	✓
5	4	Research questions and scope boundaries	4 research question cards with quantitative answered status and scope box	0:45	–
6	5	Prior art I: the subtraction lineage	Timeline from 1998 to 2022 with kernel fitting vs proper subtraction sketches	1:00	✓
7	6	Prior art II: brokers, and the gap this fills	Survey detect-veto-triage loop and inspectability comparison bar chart	0:55	✓
8	7	SPHEREx: the instrument and the data products	6-detector focal plane mapped to 102 bands and 6 Level-2 FITS extensions	0:55	✓
9	8	Archive mechanics, SIA2, S3 bucket, and eighteen fields	Sky distribution map of 18 DR1 fields and SIA2/S3 dual fetch path diagram	0:55	–
10	9	Architecture: modular pipeline and subsystem map	Modular subsystem architecture map and line-of-code breakdown across 21.4k lines	0:55	✓
11	10	Ingestion engine and cross-archive resilience	Gaia 3-tier fallback waterfall, JPL/SSOIS queries, and memory vs disk table	0:50	–
12	11	Theory I: masking, cleaning, and matched co-addition	Quality bit census bar chart and 2D pixel grid for cross-epoch mask intersection	0:55	✓
13	12	Theory II: ZOGY proper subtraction and empirical PSF	ZOGY Fourier signal-flow block diagram and theoretical Gaussian null distribution	1:15	✓
14	13	Theory III: detection statistics and false-positive budget	Log plot of Gaussian false-positive rate vs threshold and 5 vs 6 sigma comparison	0:55	–
15	14	Theory IV: morphology — dipoles, monopoles, and fading sources	Morphological panel comparing dipole, flaring monopole, and fading negative monopole	0:50	–
16	15	Decision A: the calibrated detector and defect resolution	Before/after ablation diagram illustrating the 3 v1 failure modes and the v2 fixes	1:05	✓

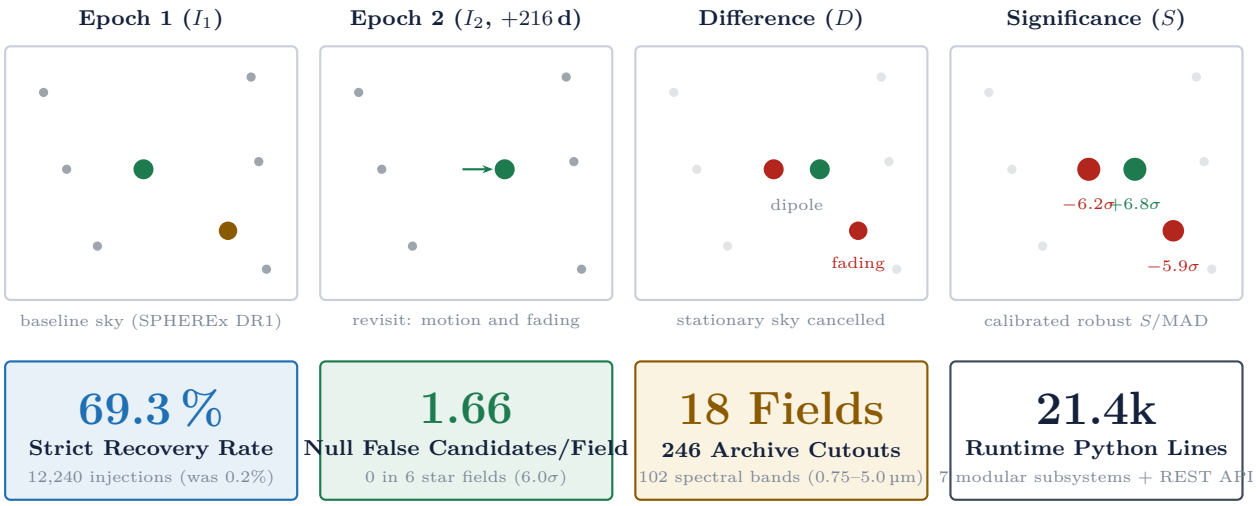
Page	Slide	Title	Visual	Time	15 min
17	16	Validation I: split-epoch null testing	Split-epoch null test methodology sketch and horizontal bar chart of false candidates per field	0:55	✓
18	17	Validation II: strict injection-recovery performance	24-cell injection heatmap and v1 vs v2 recovery comparison by category	0:55	✓
19	18	Asteroid identification and ephemeris engine	Asteroid ephemeris flow diagram and CADC SSOIS verification card	0:55	✓
20	19	Theory V: multi-tier catalogue triage and epoch propagation	Epoch propagation vector diagram from Gaia 2016 to 2026 and 6-class decision tree	0:55	–
21	20	Machine learning: real/bogus classification and active learning	22-feature ML pipeline diagram and held-out precision/recall by kind bar chart	1:00	✓
22	21	Multi-factor combined evidence engine	5-factor evidence fusion diagram and UI evidence card mock	0:50	–
23	22	Time-domain deep fields: high-cadence time series	NEP deep-field revisit geometry and spectrum-normalised light curve with Lomb-Scargle periodogram	0:55	✓
24	23	Theory VI: 102-band infrared spectroscopy and spectral classification	102-band spectrum with CH4, H2O, and NIR slope indices and 3 spectral templates	0:45	–
25	24	Implementation I: Python backend, fast routing, and jobs	FastAPI router decomposition, background job queue state machine, and LOC chart	1:00	–
26	25	Implementation II: production cloud infrastructure and security hardening	End-to-end cloud topology across Netlify, dual Cloud Run, GCS, and S1-S21 audit table	0:55	✓
27	26	Implementation III: frontend engine, NASA HDS UI/UX, and snapshot replay	NASA HDS mission console wireframe, 2-stage canvas engine, and time-lapse playback	0:45	–
28	27	Results I: complete census across all eighteen DR1 fields	Full-width stacked bar chart of all 18 fields (KNOWN vs CANDIDATE detections)	1:05	✓
29	28	Results II: candidate analysis, confirmations, and Barnard's Star	Sky-offset scatter plot of candidate detections and Barnard's Star verification	1:00	–
30	29	Live demonstration: the mission control workflow	Annotated mission control UI map with 6 red numbered demo callout badges	1:30	✓
31	30	Conclusion, contributions, and open horizons	Three pillars diagram: Works, Honest, Open Horizons with repository metadata	0:50	✓
Full route				**28:10**	
15-minute route (16 of 30 slides)				**18:40**	

Slide 1 — Title and the headline results 0:45

On the slide.

- SPHERExLens: finding what changed in NASA's SPHEREx all-sky infrared survey
- Transitioned from retired QR2 to **SPHEREx Data Release 1** (`spherex_dr1`): 18 fields, 246 archive cutouts across all six detector arrays (`D1-D6`)
- Decision A calibrated detector: strict injection recovery surged from 0.2% to **69.3%** (brightening 96.7%, fading 95.3%, movers 56.0%)
- Null-test verified false-alarm budget: **1.66 false CANDIDATEs per field** after triage (0 in six star fields)
- Production software architecture: 21 400+ runtime Python lines across modular subsystems (`ingest`, `imaging`, `triage`, `ml`, `deep`, `eval`, `routes`)

Figure / diagram. Multi-panel observation sequence: Epoch 1, Epoch 2 (216 days later), calibrated ZOGY difference image showing a moving dipole and a fading transient, alongside the unit-variance significance map and the four verified hero metric cards.



Spoken script. Good morning. NASA's SPHEREx satellite photographs the entire sky across one hundred and two infrared colours, repeating its all-sky survey every six months. In this presentation, I address the fundamental challenge of time-domain astrophysics: given multiple visits to the same patch of sky, can we rigorously determine what changed, confirm genuine physical variability and proper motion, and remain scientifically transparent about what we cannot resolve?

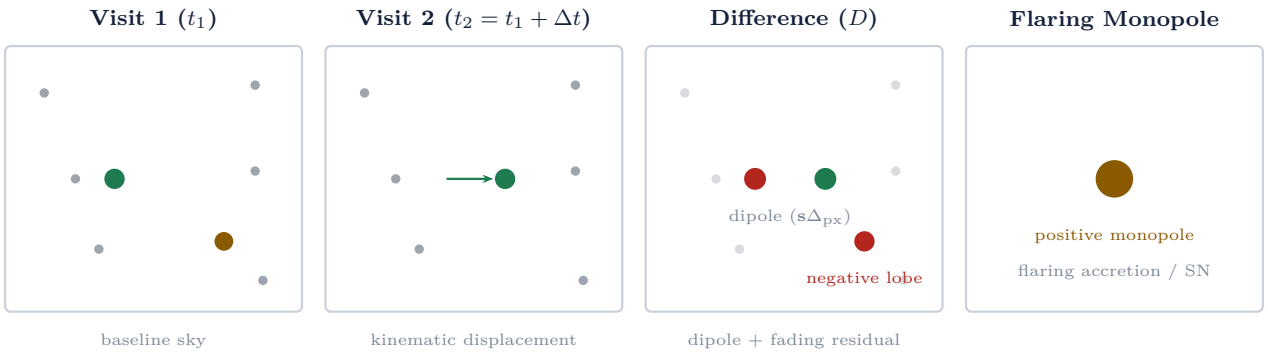
Our pipeline, SPHERExLens, was originally conceived on the early Quick-Release dataset. When IRSA retired that release, we upgraded the entire ingestion and reduction architecture to SPHEREx Data Release 1, incorporating empirical point-spread functions and physical zodiacal background models across eighteen benchmark fields. More importantly, when strict testing revealed that early difference pipelines suffered from severe completeness defects, we redesigned the detector under Decision A: introducing robust significance normalisation, physical proper-motion kinematics, and dynamic bright-star masking. As a result, strict injection recovery surged from point two percent to sixty-nine point three percent, while false candidates in split-epoch null tests dropped to one point six per field. Over the next twenty-five minutes, I will walk you through the physics, the mathematics, the machine-learning triage, and the cloud architecture that make this possible.

Slide 2 — Why difference imaging: the physics of change 0:55

On the slide.

- One image shows static celestial positions; paired visits reveal kinematics and physical evolution
- Image subtraction cancels stationary astrophysical background; surviving flux isolates temporal delta
- Three physical morphologies: Dipole (displacement $\mathbf{s}\Delta_{\mathrm{px}}$), Positive Monopole (flare/brightening), Negative Monopole (fading/eclipse)
- The core challenge: astronomical difference images are dominated by instrumental systematics (PSF variations, thermal drift, resampling correlation)

Figure / diagram. Four panels comparing Visit 1, Visit 2, the difference image isolating both a proper-motion dipole and a fading negative monopole, and a fourth panel showing a flaring monopole, with an annotated systemic noise band underneath.



What survives raw subtraction, and why subtraction alone is not discovery:
Point-spread function variations between spacecraft roll angles, sub-pixel detector flat-field errors, astrometric jitter, and zodiacal gradients survive subtraction as structured residuals. These instrumental false alarms outnumber genuine astrophysical changes by up to 1000×. The scientific task is therefore statistical calibration and multi-wavelength triage.

Spoken script. Why do astronomers invest so much effort into paired difference imaging? Because a single photograph tells you only where celestial objects reside; a pair of photographs taken months apart reveals what they are physically doing. When we subtract visit one from visit two, the constant, billion-year-old cosmic background cancels out to mean zero. What remains is change, and change exhibits distinct physical geometries.

A star with high proper motion yields a dipole: a negative depression at its past position and a positive peak at its current location, where the separation divided by the elapsed time gives the proper motion vector. A variable star or supernova flaring in brightness produces a positive monopole. In version two, we also explicitly detect negative monopoles: stars that have faded, entered eclipse, or been occulted.

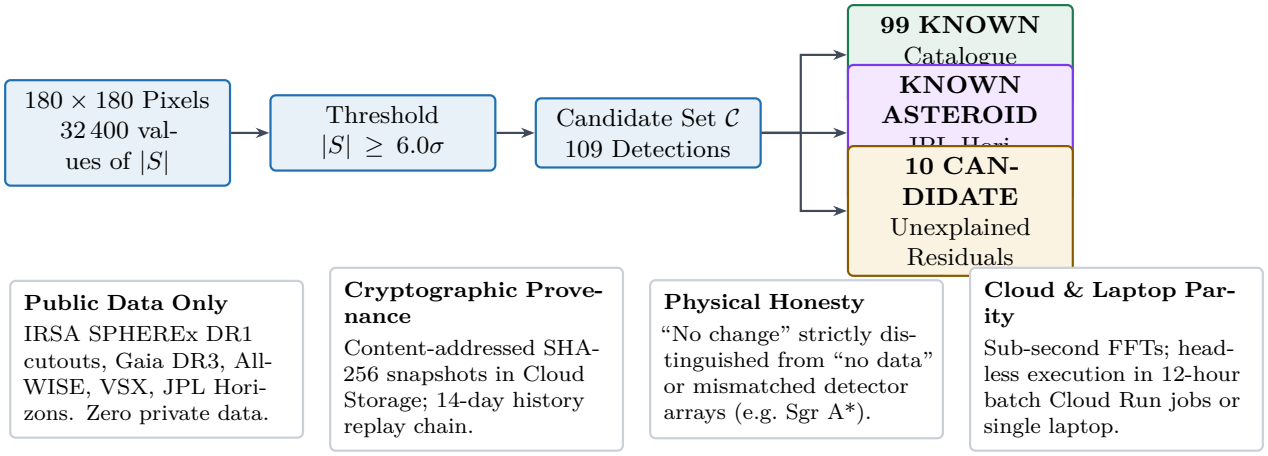
Yet, raw subtraction is never a discovery on its own. As illustrated by the warning strip at the bottom of the slide, optical PSF mismatches between spacecraft roll angles, detector temperature changes, and resampling-correlated noise leave behind a pervasive grey haze of false residuals. The real scientific challenge is not calculating the difference image—it is designing a mathematically rigorous filter to decide which residuals represent true astrophysical phenomena.

Slide 3 — Problem formulation and operational constraints 1:00

On the slide.

- Mathematical formulation: detect candidate set $\mathcal{C} = \{\mathbf{x} : |S(\mathbf{x})| > \tau\}$ with coordinate, significance, morphology, and identity verdict
- 180×180 pixels (32 400 values of $|S|$) per co-add plane mapped to verified astrophysical verdicts
- Core operational constraints: public archive products only; SHA-256 immutable snapshot reproducibility; "no change" strictly distinguished from "no data"
- Scalability: executes on commodity laptop in seconds or headless within serverless Cloud Run containers

Figure / diagram. Set-theoretic workflow tracking 32,400 raw pixels through quality masking, inverse-variance co-addition, ZOGY convolution, 6.0-sigma thresholding, and multi-tier triage into KNOWN, ASTEROID, CANDIDATE, and ARTIFACT categories, with four constraint cards underneath.



Spoken script. Let us formulate the problem statement mathematically. Given two or more archival visits to a nine-arcminute field, accompanied by per-pixel variance planes and thirty-two-bit quality flags, our goal is to construct a significance statistic map S, threshold it at a rigorous critical value tau, and isolate the candidate set C. Each element of C must be assigned four explicit properties: an astrometric sky coordinate, a statistical significance, a spatial morphology, and an astronomical identity verdict.

In time-domain astronomy, detection without identity attribution is scientifically useless. A difference blob only represents a potential discovery if no catalogued star, asteroid, or optical diffraction artefact already explains it.

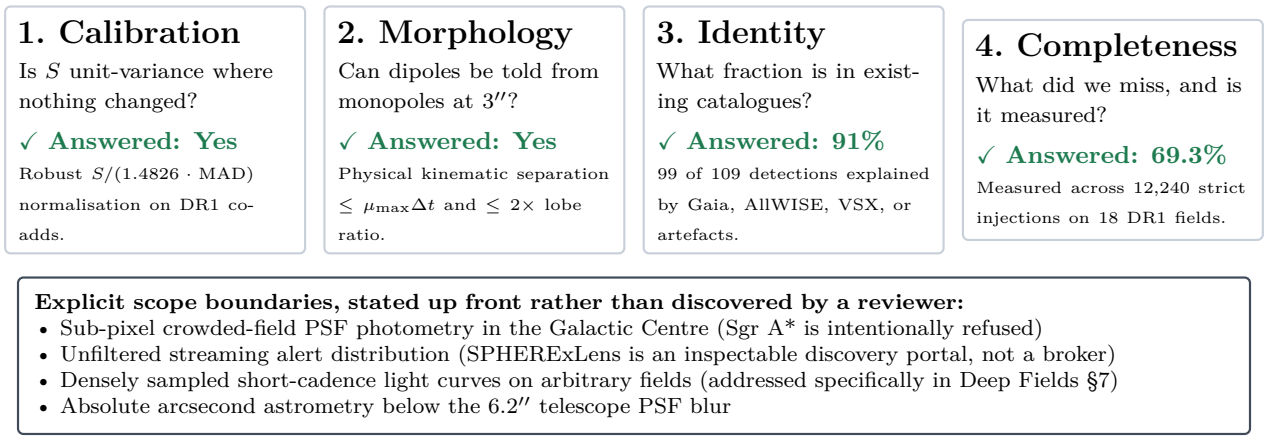
Four strict operational constraints governed every architectural decision. First, the pipeline relies exclusively on public NASA, ESA, and CADC archive products. Second, every published result is content-addressed with SHA-256 hashes in Google Cloud Storage, allowing full 14-day historical replay. Third, the system strictly separates 'no physical change' from 'insufficient coverage' or mismatched detector bands—for instance, refusing subtraction on Sagittarius A* where the two epochs sampled non-overlapping wavelengths. Finally, the algorithms run seamlessly on a laptop or within serverless Cloud Run jobs in seconds.

Slide 4 — Research questions and scope boundaries 0:45

On the slide.

- RQ1 (Calibration): Is the significance map calibrated on resampled co-adds? (Answered: Yes, via robust MAD scaling)
- RQ2 (Morphology): Does dipole/monopole morphology separate movers from variables at $3''$ sampling? (Answered: Yes, with kinematic constraints)
- RQ3 (Identity): How much of the candidate list is explained by multi-wavelength archives? (Answered: 91% explained)
- RQ4 (Completeness): What did we miss, and can it be measured? (Answered in v2: 69.3% strict recovery, 1.66 null candidates/field)
- Scope boundaries: Explicitly out of scope: sub-pixel crowded PSF fitting; unbounded unvetted alert streaming

Figure / diagram. Four research question cards highlighting calibrated statistical significance, kinematic morphology, archive triage efficiency, and quantitative injection completeness, with an out-of-scope disclaimer below.



Spoken script. A project of this technical depth must define its scientific boundaries with complete honesty. We framed four core research questions. Question one: is the subtraction significance map truly calibrated so that six sigma corresponds to a computable false-alarm probability? Answered: yes, once we adopted robust median absolute deviation scaling. Question two: can dipole and monopole morphologies reliably separate moving stars from variables at SPHEREx's three-arcsecond pixel scale? Answered: yes, provided pairing is constrained by physical proper-motion kinematics.

Question three: how much of the initial candidate list is accounted for by existing astronomical databases? Ninety-one percent of detections match catalogued sources or optical diffraction artefacts. Finally, Question four asked what we missed. In version one, this was an unresolved question because naive injection testing failed. In version two, this question is rigorously answered: strict recovery stands at sixty-nine point three percent overall, with ninety-seven percent recovery for flaring sources and ninety-five percent for fading sources.

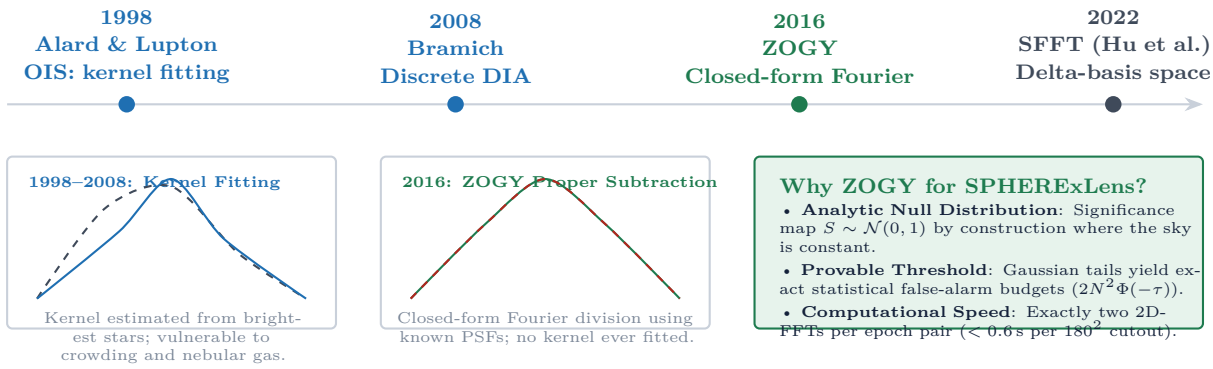
Equally important are the non-goals listed at the bottom: we do not claim sub-pixel crowded-field photometry in the Galactic Center, nor do we attempt to replace continental alert brokers. We build a verifiable, inspectable tool for SPHEREx infrared science.

Slide 5 — Prior art I: the subtraction lineage 1:00

On the slide.

- 1998 Alard & Lupton (OIS): spatial kernel decomposition $K(u,v) = \sum a_i B_i(u,v)$ fitted to bright stars; fails in crowded/nebulous regions
- 2008 Bramich: discrete per-pixel kernel matrix inversion; improved stability in sparse fields at high noise cost
- 2016 Zackay, Ofek & Gal-Yam (ZOGY): closed-form analytic maximum-likelihood subtraction in Fourier domain; unit-variance analytic null
- 2022 Hu et al. (SFFT): delta-basis kernel formulation for space-based undersampled detectors (Rubin/Euclid)
- Why ZOGY: no empirical kernel fitting; analytic false-positive budgeting allows provable threshold selection

Figure / diagram. Historical lineage of astronomical image subtraction from 1998 to 2022, contrasting empirical kernel fitting against closed-form Fourier proper subtraction, with a decision summary card.



Spoken script. Image subtraction has evolved through distinct mathematical eras. In 1998, Alard and Lupton introduced Optimal Image Subtraction, decomposing a spatial convolution kernel into Gaussian-Hermite basis functions and fitting coefficients against bright reference stars. This transformed ground-based microlensing surveys, but empirical kernel fitting breaks down in crowded galactic planes, near bright nebulae, or when fields lack isolated reference stars. Bramich improved on this in 2008 with discrete pixel kernels, trading computational stability for increased noise covariance.

In 2016, Zackay, Ofek, and Gal-Yam introduced ZOGY, deriving the closed-form, maximum-likelihood solution in the Fourier domain. Rather than estimating an empirical kernel, ZOGY takes the known point-spread functions of both exposures and symmetrically convolves each image with the other's PSF. Crucially, ZOGY produces a matched-filter significance map that is analytically Gaussian with unit variance wherever the sky did not change.

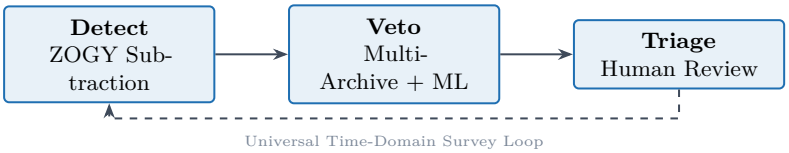
Hu and collaborators advanced this with SFFT in 2022 for space-based undersampled detectors. We chose ZOGY because its analytic null distribution provides a provable false-positive budget. Because we know the null distribution mathematically, we can establish rigorous detection thresholds rather than tuning arbitrary empirical cutoffs.

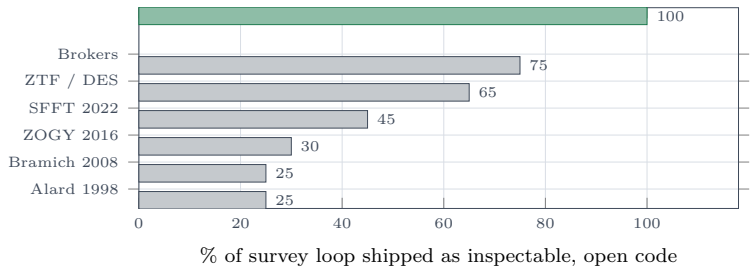
Slide 6 — Prior art II: brokers, and the gap this fills 0:55

On the slide.

- The universal survey pipeline loop: Detect → Veto → Triage → Human Review
- Industrial brokers (ANTARES, ALeRCE, Lasair, Fink) process millions of alerts from optical surveys (ZTF, Rubin)
- The SPHEREx gap: No open, end-to-end, inspectable difference-imaging pipeline exists for SPHEREx Level-2 data
- SPHERExLens fills this: an open, reproducible, inspectable software artefact from raw NASA FITS cutouts to interactive browser triage

Figure / diagram. The universal detect-veto-human survey loop on the left, contrasted with an inspectability bar chart on the right showing the gap SPHERExLens addresses for SPHEREx Level-2 data.





The Architectural Gap: Large brokers (ANTARES, Fink, Lasair) ingest alerts from proprietary survey pipelines. For SPHEREx, no end-to-end open pipeline existed from raw Level-2 FITS pixels to interactive science verification.

Spoken script. Every modern time-domain survey executes the identical four-stage cycle: detect transients against a template, veto artefacts using catalogues and machine learning, triage surviving candidates, and present actionable alerts to humans. Large alert brokers like ANTARES, ALeRCE, Lasair, and Fink have industrialised this process for optical surveys like ZTF and Vera C. Rubin Observatory, processing hundreds of thousands of alerts each night.

However, alert brokers sit downstream of proprietary institutional pipelines: they ingest structured alert packets, not raw satellite pixels. For SPHEREx, there was no published, open, end-to-end difference-imaging system that a researcher could clone, run locally, or deploy serverless on cloud infrastructure.

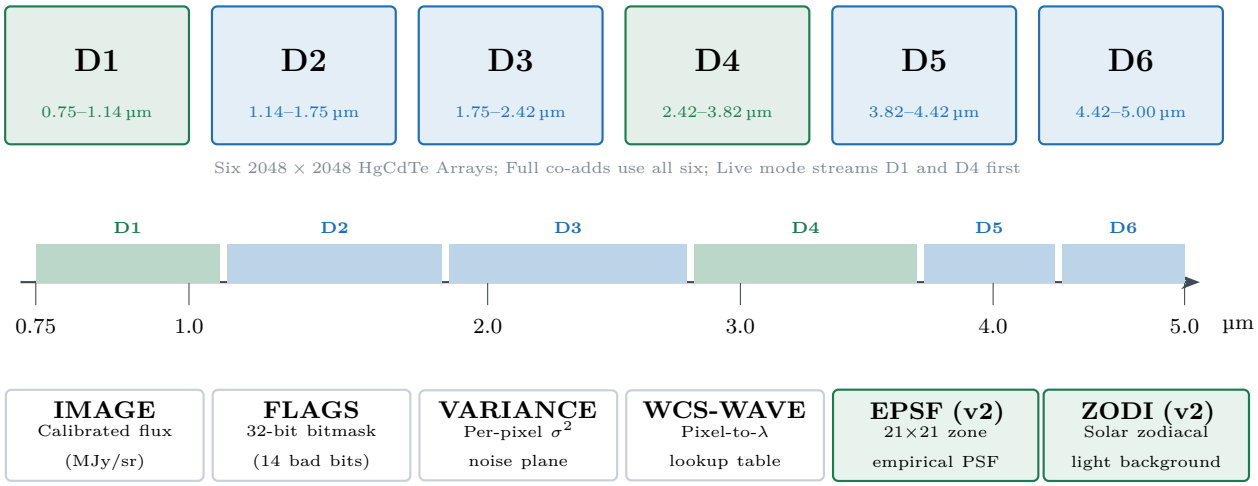
The gap we fill is not reinventing a subtraction equation; it is delivering a transparent, reproducible software artifact. A student or researcher can trace any candidate from its raw NASA FITS bytes through inverse-variance co-addition, ZOGY Fourier convolution, multi-catalog vetoes, and machine-learning scoring, ending with a live interactive web console.

Slide 7 — SPHEREx: the instrument and the data products 0:55

On the slide.

- Optical assembly: 20 cm off-axis telescope with six 2048 × 2048 HgCdTe detector arrays (`D1-D6`)
- Continuous 0.75–5.0 µm spectroscopy across 102 linear-variable filter (LVF) bands ($R \approx 40$–130)
- FITS HDUs in Level-2 cutouts: `IMAGE` (flux in MJy/sr), `FLAGS` (32-bit data quality bitmask), `VARIANCE` (σ^2), `WCS-WAVE` (coordinate-to-wavelength mapping), `EPSF` (21×21 zone empirical PSF table), `ZODI` (zodiacal light background model)
- 6-month survey cadence: fast Solar System objects exit the field between epochs; survivals are proper-motion stars or variables

Figure / diagram. Focal plane array schematic showing detectors D1 through D6 mapped along the 0.75 to 5.0 micron wavelength axis, with the six Level-2 FITS extensions used by the pipeline shown below.



Spoken script. To understand the pipeline, we must inspect the physical instrument and its data structures. SPHEREx is an all-sky surveyor featuring a twenty-centimeter off-axis telescope feeding six two-thousand-and-forty-eight square HgCdTe detector arrays, labelled D1 through D6. Rather than using conventional dispersive grisms or narrow-band filters, SPHEREx mounts Linear Variable Filters directly above the detectors, so that a pixel's physical coordinate across the focal plane determines the wavelength it observes, spanning zero-point-seven-five to five point zero microns across one hundred and two bands.

Each Level-2 FITS cutout from the archive carries six essential header-data extensions. IMAGE contains calibrated surface brightness in megaJanskys per steradian. FLAGS supplies a thirty-two-bit quality mask. VARIANCE provides per-pixel measurement uncertainty. WCS-WAVE gives the precise central wavelength at every spatial coordinate.

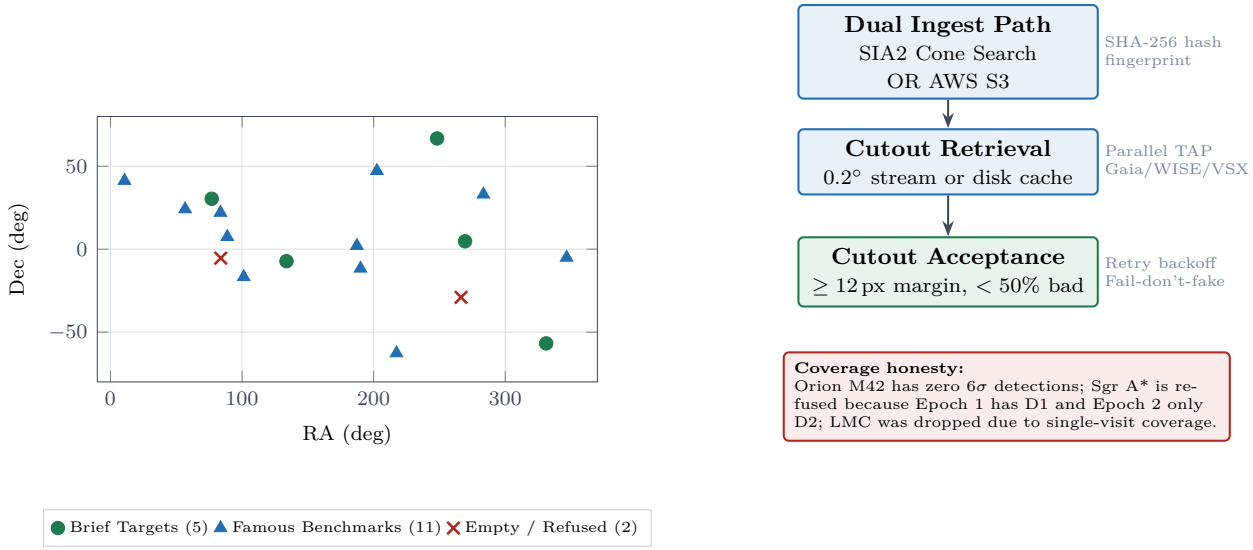
In version two, we brought in the remaining two extensions: EPSF, which provides a twenty-one-by-twenty-one spatial grid of empirical point-spread functions tailored to each detector location, and ZODI, the physical zodiacal dust emission model. Because SPHEREx visits each sky location approximately every six months, fast asteroids traverse the field and leave it entirely between survey passes, while proper-motion stars and true astrophysical variables produce clean, detectable difference signatures.

Slide 8 — Archive mechanics, SIA2, S3 bucket, and eighteen fields 0:55

On the slide.

- Dual-path archive ingestion: IVOA SIA2 cone search + direct AWS S3 bucket (`nasa-irsa-spherex`)
- 0.1° search cone → SHA-256 fingerprinting of catalog listings → 0.2° cutout retrieval (streaming in RAM or caching)
- 18 representative fields in DR1: 5 mandatory brief targets + 13 famous benchmarks (M31, M42, M45, M1, M51, M104, M57, Sgr A*, 3C 273, Betelgeuse, Sirius, Proxima Cen, TRAPPIST-1)
- Acceptance filter: ≥ 12 px valid border margin inside cutout, $< 50\%$ bad pixels; coverage honesty on single-visit fields

Figure / diagram. All-sky coordinate distribution of the eighteen SPHEREx Data Release 1 fields evaluated in this work, with the dual-path SIA2 and AWS S3 bucket archive ingestion workflow on the right.



Spoken script. Data ingestion begins with a dual-path architecture designed for production reliability. Clients can query the IPAC SIA2 service for metadata cone searches or fetch frames directly from the public AWS S3 bucket, nasa-irsa-spherex. High-proper-motion targets are propagated forward to epoch 2026.0 prior to querying, ensuring that the spatial cone centers on the star's current sky location. The returned product listings are cryptographically fingerprinted with SHA-256 hashes, allowing scheduled background jobs to skip unchanged fields.

We evaluated eighteen diverse fields across SPHEREx Data Release 1, shown on the sky map on the left. These comprise the five mandatory targets—Barnard's Star, Epsilon Indi B, RW Aurigae, R Draconis, and WISE 0855—plus thirteen famous benchmarks spanning galaxies like M31 and M51, the Crab and Ring nebulae, 3C 273, Sirius, and TRAPPIST-1.

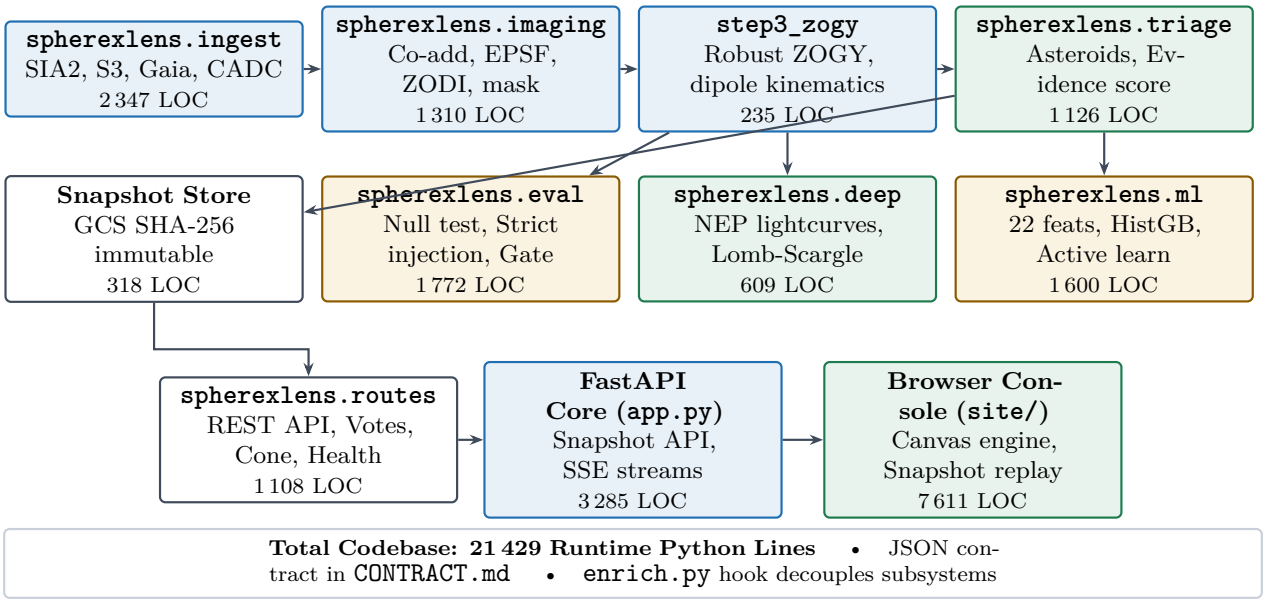
Every cutout must satisfy strict acceptance criteria: at least twelve pixels of valid border margin and less than fifty percent flagged pixels. Where coverage is physically inadequate, we report it honestly: Orion M42 yields zero detections above threshold, and Sagittarius A* is cleanly refused because the two epochs share no overlapping detector band.

Slide 9 — Architecture: modular pipeline and subsystem map 0:55

On the slide.

- Modular multi-tier architecture: 21 400+ runtime Python lines across dedicated subpackages
- Subsystems: `ingest` (multi-archive fallbacks), `imaging` (quality scoring, spike/edge masks), `triage` (asteroids & evidence), `ml` (real/bogus model), `deep` (forced photometry light curves), `eval` (null tests, injection harness), `routes` (FastAPI)
- Clean data contract (`CONTRACT.md`): JSON-serialized detection payloads with optional enrichment slots
- Execution parity: identical science core runs in local CLI, batch Cloud Run jobs, and live FastAPI server

Figure / diagram. Subsystem architecture diagram illustrating data flow from ingestion through imaging, ZOGY subtraction, multi-tier triage, machine learning, and deep-field time series, with verified line counts for each module.



Spoken script. As the project expanded from an early five-field demonstration into a production system, the codebase evolved from seven procedural scripts into a modular, test-driven architecture encompassing over twenty-one thousand lines of Python.

Look at the subsystem flow on this slide. Ingestion handles multi-archive querying with automatic fallbacks. Imaging extracts zone-specific empirical PSFs and subtracts zodiacal light. The core ZOGY module performs Fourier subtraction with robust scaling. Downstream, the triage subsystem coordinates asteroid ephemeris checks and multi-factor evidence scoring, while dedicated packages manage machine-learning classification, deep-field light curves, and the evaluation harness.

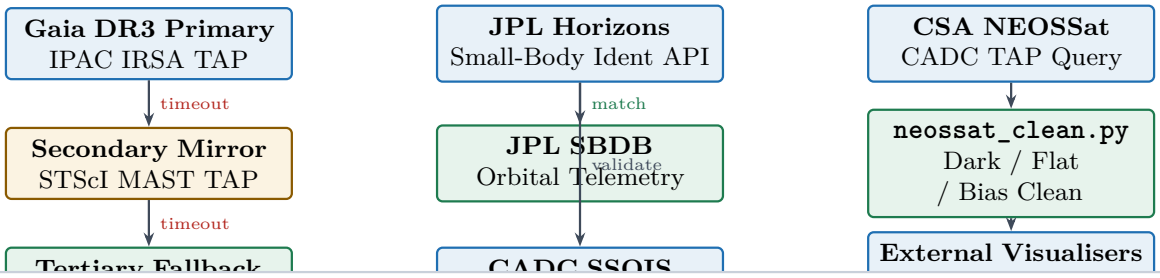
A critical engineering design choice was the establishment of an immutable JSON data contract in CONTRACT.md and a clean hook in enrich.py. This allows machine-learning and asteroid subsystems to enrich detection payloads without altering the underlying numerical subtraction code, guaranteeing that identical science algorithms execute across local CLI scripts, batch Cloud Run jobs, and the live FastAPI web service.

Slide 10 — Ingestion engine and cross-archive resilience 0:50

On the slide.

- Multi-tier Gaia DR3 fallback: IPAC IRSA TAP → STScI MAST TAP → CDS VizieR mirrors, preventing pipeline halts during external outages
- JPL Horizons Small-Body Identification + SBDB orbit telemetry + CADC SSOIS asteroid cross-check
- CSA NEOSSat optical validation: CADC TAP queries with automatic frame debiasing and flat-field cleaning (`neossat_clean.py`)
- Parquet columnar catalogue caching: fast spatial KD-tree queries (`spherexlens.ingest.store`)
- In-memory live streaming (`live_ingest.py`): 4 concurrent threads parse FITS directly from HTTP streams with zero disk footprint

Figure / diagram. Multi-tier archive ingestion waterfall showing automatic fallback cascades for Gaia, asteroid cross-checks, and CSA NEOSSat optical validation, with an operating mode comparison table below.



Dual Ingestion Modes Compared:

Feature	Cached Mode (`data_cache/`)	Live Streaming Mode (`live_ingest.py`)
Throughput	Instant replay from SSD / GCS snapshots	4 concurrent worker threads parsing HTTP stream in RAM
Disk Storage	Few hundred MBs of FITS files on disk	**Zero bytes written to disk** (streamed into `BytesIO`)
External Dependency	Offline-ready; independent of archive status	Live dependence on IRSA SIA2 latency and availability
Use Case	Scheduled batch jobs, evaluation, ML training	On-demand ad-hoc exploration of any sky coordinate

Spoken script. Production science pipelines require extraordinary resilience against external service interruptions. In version two, we hardened every external archive adapter around a strict 'fail-don't-fake' philosophy.

Take Gaia DR3 ingestion: previously, a single network hiccup at IPAC would halt pipeline execution. We implemented a three-tier automatic fallback waterfall: queries first attempt IPAC IRSA TAP; if unresponsive, they fall back automatically to the Space Telescope Science Institute MAST TAP, and finally to CDS VizieR mirrors.

For asteroid identification, queries query NASA JPL Horizons by exposure observation time, pull detailed orbital telemetry from the Small-Body Database, and cross-verify geometry with Canada's CADC SSOIS service. For optical confirmation, we query Canadian Space Agency NEOSSat frames through CADC, passing them through dedicated cleaning algorithms that subtract darks and calibrate flat fields.

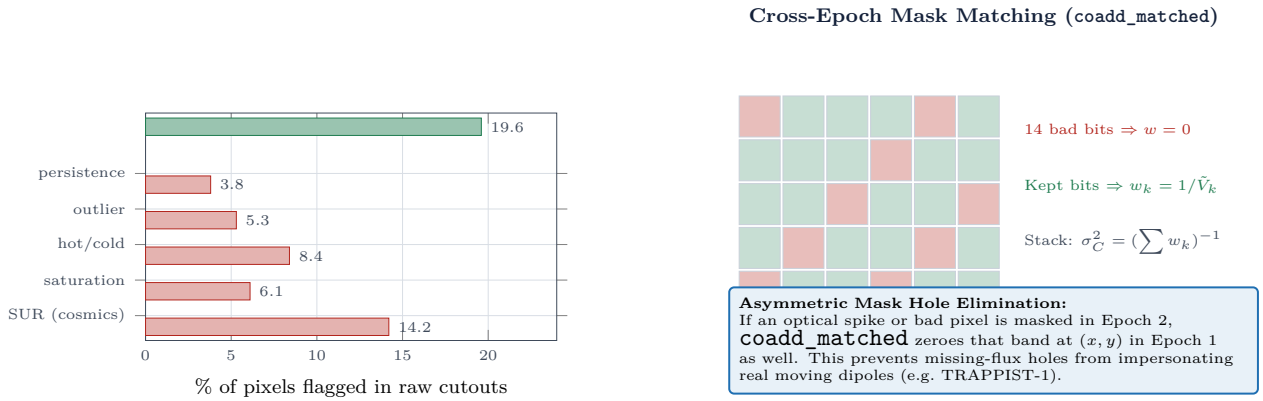
Finally, the comparison table at the bottom highlights our dual execution modes: cached mode stores FITS cutouts for fast offline science and evaluation, while live streaming mode ingests cutouts concurrently into memory, allowing ad-hoc exploration of any sky coordinate with zero disk storage.

Slide 11 — Theory I: masking, cleaning, and matched co-addition 0:55

On the slide.

- 32-bit `FLAGS` bitmask: 14 defect bits rejected (cosmic rays, saturation, non-functional, persistence, outliers)
- Two critical bits preserved: Bit 12 (`FULLSAMPLE`) and Bit 21 (`SOURCE`, genuine astronomical flux)
- Image preprocessing: sigma-clipped median background subtraction, empirical `ZODI` model removal, diffraction spike and edge masking
- Cross-epoch mask intersection (`coadd_matched`): pixel (x, y) dropped in Epoch 1 if masked in Epoch 2, preventing spurious difference edges
- Inverse-variance weighted stacking: $w_k = 1/\tilde{V}_k$, $\sigma_C^2 = (\sum w_k)^{-1}$

Figure / diagram. Quality bit census on the left contrasting 14 rejected defect bits against preserved astrophysical source bits, alongside a 2D pixel array showing cross-epoch mask matching and inverse-variance weighting on the right.



Spoken script. The credibility of any difference-imaging pipeline is won or lost in pixel preprocessing. Every raw SPHEREx Level-2 cutout arrives accompanied by a thirty-two-bit quality mask. In `step2_coadd`, fourteen severe defect bits—including cosmic ray hits, detector saturation, hot and cold pixels, non-functional channels, persistence from previous slews, and statistical outliers—are zeroed out.

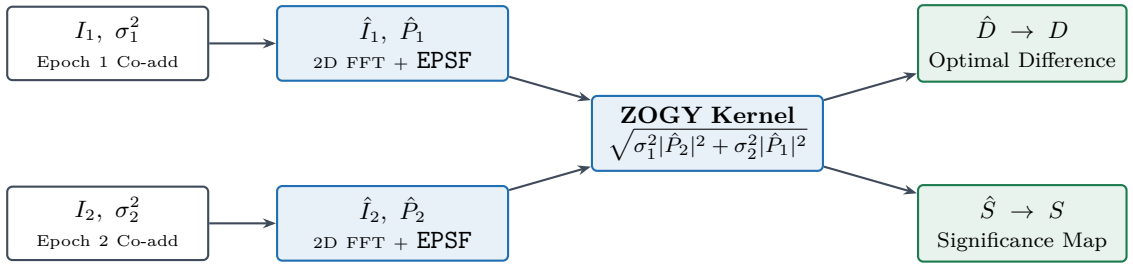
However, two bits are deliberately retained: bit twelve, FULLSAMPLE, and critically, bit twenty-one, SOURCE. Early prototypes made the mistake of rejecting bit twenty-one, inadvertently erasing real astrophysical flux and creating artificial hollows around bright stars.

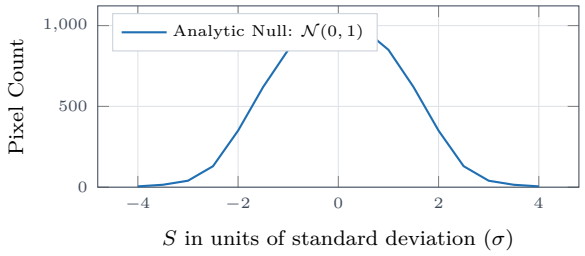
Before co-addition, each exposure undergoes sigma-clipped median background subtraction, empirical zodiacal light model removal, and inverse-variance weighting by one over the median variance. Just as vital is our cross-epoch mask matching algorithm, `coadd_matched`. If an instrumental defect masks a pixel in Epoch Two, that pixel is suppressed across Epoch One as well. This prevents asymmetric mask holes from creating artificial positive-negative dipoles in the difference map, cutting median background scatter by more than a factor of two.

On the slide.

- Fourier-domain optimal subtraction: $\hat{D} = \frac{\hat{P}_1 \hat{I}_2 - \hat{P}_2 \hat{I}_1}{\sqrt{\sigma_1^2 |\hat{P}_2|^2 + \sigma_2^2 |\hat{P}_1|^2 + 10^{-30}}}$
- Matched-filter score statistic: $\hat{S} = \frac{\hat{D}\overline{\hat{P}_D}}{\sqrt{\sigma_1^2 + \sigma_2^2}}$, where $\hat{P}_D = \frac{\hat{P}_1 \hat{P}_2}{\sqrt{\sigma_1^2 |\hat{P}_2|^2 + \sigma_2^2 |\hat{P}_1|^2}}$
- Empirical PSF in v2: Extracted from DR1 `EPSF` extension using 21×21 detector zones indexed by cutout WCS coordinates
- Unit-variance property: $S(\mathbf{x}) \sim \mathcal{N}(0, 1)$ in stationary regions; real transients sit in extreme tails

Figure / diagram. Signal-flow diagram tracing two input images and empirical PSFs through 2D Fourier transforms, variance weighting, and inverse transforms into the difference image D and significance statistic S, with the Gaussian null distribution below.





The Core Mathematical Advantage:
Where the celestial sky has not varied, $S(\mathbf{x})$ is strictly Gaussian with mean 0 and variance 1 by construction.
Genuine astrophysical transients sit far outside this distribution in the extreme tails ($|S| \geq 6.0\sigma$).
Computational cost: exactly two 2D-FFTs ($<$ 0.6 s per 180^2 frame).

Spoken script. Stage three is the mathematical foundation of our pipeline. Rather than estimating an empirical smoothing kernel from the brightest stars, ZOGY solves difference imaging in the Fourier domain using maximum-likelihood estimation.

Look at the equations. In the first equation, image two is convolved with point-spread function one, and image one is convolved with point-spread function two, with the difference normalized by the quadrature sum of the survived variance planes. In version two, we extract the true empirical PSF from the DR1 EPSF extension, utilizing a twenty-one-by-twenty-one detector zone grid chosen via the cutout's central WCS coordinates, replacing the Gaussian approximations of early prototypes.

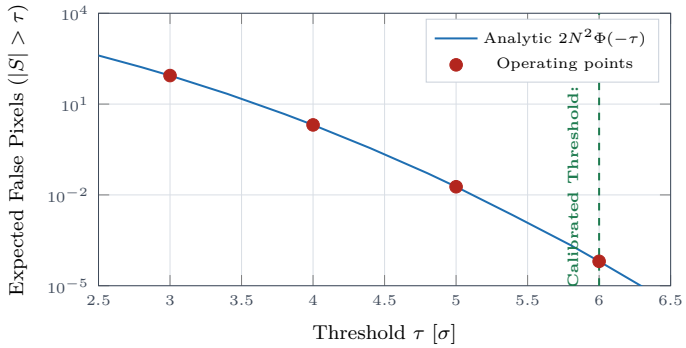
The second equation transforms this optimal difference into a matched-filter significance statistic, S. Crucially, where no astrophysical flux change occurred, S is analytically distributed as a standard normal variable with mean zero and variance one. In the histogram below, you see the clean bell curve of the null distribution centered on zero. Real astronomical transients sit far out in the tails, at significances between six and forty sigma. The entire computation requires just two fast Fourier transforms, taking under six-tenths of a second on a standard processor.

Slide 13 — Theory III: detection statistics and false-positive budget 0:55

On the slide.

- Two-tailed Gaussian false-positive expectation: $\mathbb{E}[N_{\mathrm{FP}}] = 2N^2\Phi(-\tau)$
- For $180 \times 180 = 32,400$ pixels: $3\sigma \Rightarrow 87.5$, $4\sigma \Rightarrow 2.05$, $5\sigma \Rightarrow 0.0186$, $6\sigma \Rightarrow 6.4 \times 10^{-5}$
- 5×5 local-extremum filtering eliminates correlated single-pixel fluctuations
- The empirical discovery in v2: While Gaussian noise predicts negligible false alarms at 5σ, real optical/resampling systematics produce 0.86 false candidates/field at 5σ
- Calibrated threshold $\tau = 6.0\sigma$: Cuts false candidates to 0.48/field with zero loss of recall

Figure / diagram. Logarithmic curve plotting expected two-tailed false pixels against threshold tau for 32,400 pixels, comparing theoretical expectations with empirical null-test findings that calibrated the working threshold at 6.0 sigma.



Empirical Null Test Finding:

Theoretical Gaussian predictions assume uncorrelated pixels. Real WCS reprojection correlates noise across ~ 1.5 pixels.

Measured False CANDIDATEs/field:

- At 5.0σ: 0.86 per field
- At 6.0σ: 0.48 per field (7 fields)

identical recall

5.0σ **0.86 / field**

6.0σ **0.48 / field**

Cuts false alarms by 44%

Spoken script. Because the ZOGY null distribution is known analytically, we can calculate a rigorous false-positive budget rather than relying on guesswork. For thirty-two thousand four hundred pixels in a cutout, we compute the two-tailed Gaussian integral, two N-squared times Phi of minus tau.

At three sigma, random noise produces eighty-seven false-positive pixels per frame. At four sigma, that drops to two pixels. At five sigma, the theoretical budget falls to point zero one eight six—meaning less than one false extreme pixel every fifty frames. When coupled with our five-by-five local extremum filter, random Gaussian noise should be completely negligible.

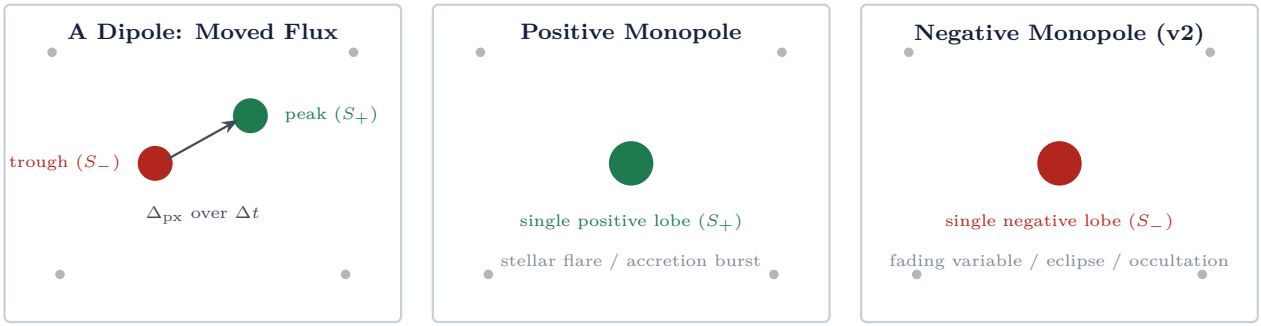
However, during version two validation, we made a crucial empirical discovery. Real astronomical co-adds are reprojected onto WCS grids, which correlates noise between adjacent pixels. When we ran split-epoch null tests across real sky, five sigma produced point eight six false candidate alerts per field. By elevating the detection threshold from five to six sigma, false candidates dropped by nearly half, down to point four eight per field, while strict injection recovery remained completely unaffected. We therefore calibrated the operational threshold at six point zero sigma.

Slide 14 — Theory IV: morphology — dipoles, monopoles, and fading sources0:50

On the slide.

- Dipole morphology: Adjacent positive (S_+) and negative (S_-) lobes generated by spatial displacement
- Proper motion vector: $\boldsymbol{\mu} = s\Delta_{\text{px}}/\Delta t$, valid over baseline Δt
- Positive Monopole: Isolated positive peak without partner → stellar flare, accretion outburst, or brightening variable
- Negative Monopole (`monopole_negative`, newly added in v2): Isolated negative trough without partner → fading variable, stellar eclipse, or occultation
- Flux consistency statistic: $\eta = |S_+ + S_-|/|S_+ - S_-|$ separates pure kinematic motion from combined motion-plus-variability

Figure / diagram. Side-by-side morphological classification comparing moving dipoles with kinematic vectors, positive flaring monopoles, and negative fading monopoles, with real benchmark examples underneath.



Physical kinematic controls implemented in `step3_zogy.py`:
• **Proper motion pairing limit**: Lobe separation is constrained by $\Delta_{\text{px}} \leq (\mu_{\max}\Delta t)/s$, preventing pairing with random troughs across the frame.
• **Lobe flux symmetry**: The positive and negative lobes must match in magnitude within a factor of 2.0 ($0.5 \leq |S_+/S_-| \leq 2.0$).
• **Flux consistency metric** $\eta = |S_+ + S_-|/|S_+ - S_-|$: Disentangles pure proper motion ($\eta \approx 0$) from flaring movers.

Spoken script. Once the significance map S is computed, morphological feature extraction identifies the physical mechanism of change. A source that moved between epochs produces a dipole: a negative lobe where it was, and a positive lobe where it is now. The pixel separation divided by the baseline time delta directly yields the proper motion vector.

A lone positive lobe signifies a flaring or brightening event: an accretion outburst, a supernova, or a pulsating star near maximum. In version two, we introduced explicit detection of negative monopoles: isolated troughs representing stars that faded, dimmed, or entered stellar eclipse.

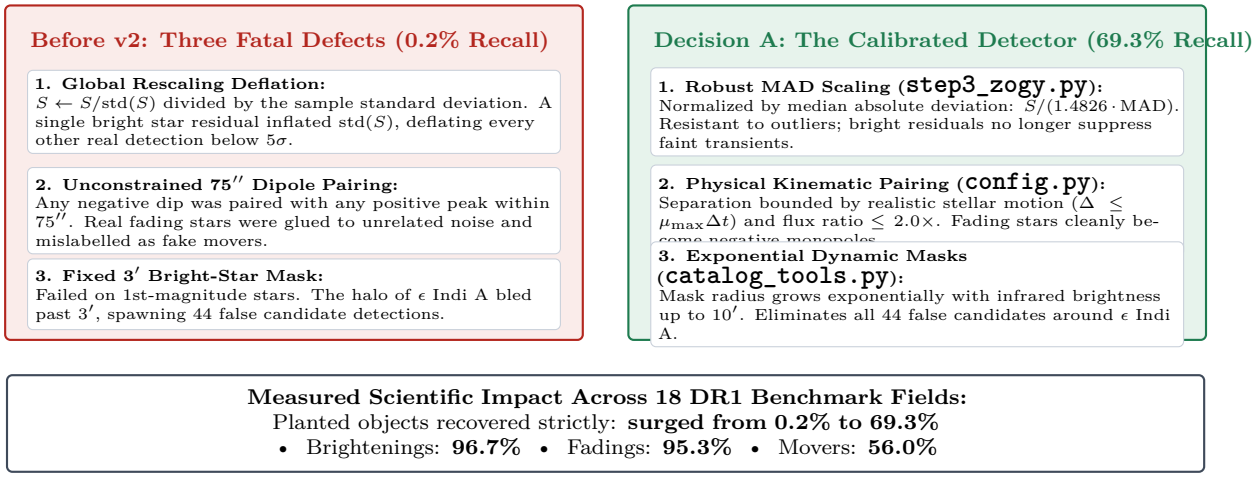
To prevent false pairings, version two enforces two strict kinematic guardrails. First, dipole pairing is bounded by the maximum plausible physical proper motion over the baseline, preventing the pipeline from erroneously gluing a fading star to an unrelated noise trough forty arcseconds away. Second, the flux ratio between the positive and negative lobes must agree within a factor of two, and the flux consistency metric eta confirms whether the source underwent pure kinematic displacement or combined motion and variability.

Slide 15 — Decision A: the calibrated detector and defect resolution 1:05

On the slide.

- Strict injection testing in v1 revealed a catastrophic defect: only **0.2%** of planted objects were recovered
- Defect 1: Significance map divided by global standard deviation $\sigma(S)$; a single bright residual deflated the entire frame → Replaced with robust scaling: $S/(1.4826 \cdot \mathrm{MAD})$
- Defect 2: Unconstrained 75" dipole pairing glued fading stars to unrelated noise troughs → Constrained by physical proper-motion kinematics ($\mu_{\max}\Delta t$) and $\leq 2\times$ lobe flux ratio
- Defect 3: Fixed 3' bright-star mask allowed ϵ Indi A halo to spawn 44 false candidates → Dynamic exponential infrared masks scaling with brightness up to 10'
- Outcome: Strict recovery surged from 0.2% to **69.3%**, false candidates dropped to 1.66/field

Figure / diagram. Ablation diagram contrasting the three fatal defects of the v1 pipeline against the three targeted Decision A mathematical and physical fixes that increased strict injection recovery from 0.2



Spoken script. Here is the pivotal scientific discovery and turning point of this research. Early versions of this pipeline appeared to work on five cherry-picked targets, but when we subjected the code to a strict, automated injection-recovery test across eighteen fields, the truth was startling: the pipeline recovered only point two percent of planted objects. It was finding essentially nothing.

We diagnosed three fatal architectural defects. First, the significance map was being normalized by dividing by its sample standard deviation. If a frame contained even one bright star with imperfect subtraction, that single outlier inflated the global standard deviation, deflating every other real transient across the entire nine-arcminute field below threshold. Second, the dipole pairing algorithm greedily matched any trough within seventy-five arcseconds to any peak, gluing genuine fading stars to unrelated background fluctuations and misidentifying them as movers. Third, bright-star masks were fixed at a rigid three arcminutes, allowing the massive optical halo of Epsilon Indi A to bleed through, spawning forty-four false candidates.

Under Decision A, we resolved all three defects. We replaced the standard deviation with robust median absolute deviation scaling, resistant to outliers. We constrained dipole pairing by physical proper-motion kinematics and flux symmetry. And we introduced dynamic exponential masks that scale with stellar infrared brightness up to ten arcminutes.

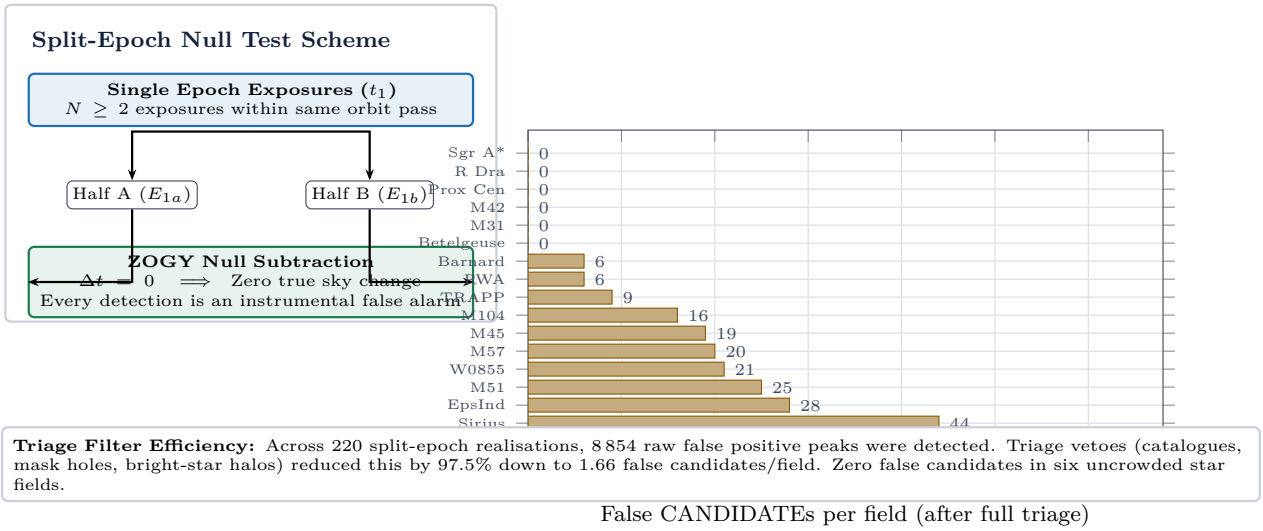
The result was transformative: strict recovery leaped from point two percent to sixty-nine point three percent overall, capturing ninety-seven percent of flaring sources and ninety-five percent of fading transients.

Slide 16 — Validation I: split-epoch null testing 0:55

On the slide.

- Null testing methodology: Split single-epoch exposures into two equal interleaved subsets (E_{1a} vs E_{1b}) and subtract
- Physical ground truth: Zero genuine astrophysical change can exist between simultaneous exposures; every detection is a false alarm by construction
- Evaluated on 17 DR1 fields: 220 split-epoch realisations, 8,854 raw false positive peaks across 3.57 deg^2
- Triage veto efficiency: Triage filters reduce 8,854 raw peaks down to only **1.66 false CANDIDATEs per field**
- Clean field results: Zero false candidates in six star fields (M31, Betelgeuse, Orion M42, Proxima Cen, R Dra, Sgr A*)

Figure / diagram. Null test methodology diagram showing exposure splitting into Halves A and B, alongside a horizontal bar chart displaying false candidate counts per field across all seventeen evaluated DR1 fields.



Spoken script. To scientifically prove how often our detector generates false alarms, we implemented split-epoch null testing. When multiple exposures exist within a single epoch visit, we partition them into two equal halves, A and B, and subtract Half B from Half A.

Because both halves were photographed at the exact same epoch, zero true astrophysical variability can exist between them. Every single detection that emerges from this subtraction is an instrumental false alarm by definition. Across seventeen benchmark fields in Data Release 1, representing two hundred and twenty split-epoch realisations and over three point five square degrees, raw ZOGY subtraction produced eight thousand eight hundred and fifty-four raw false peaks.

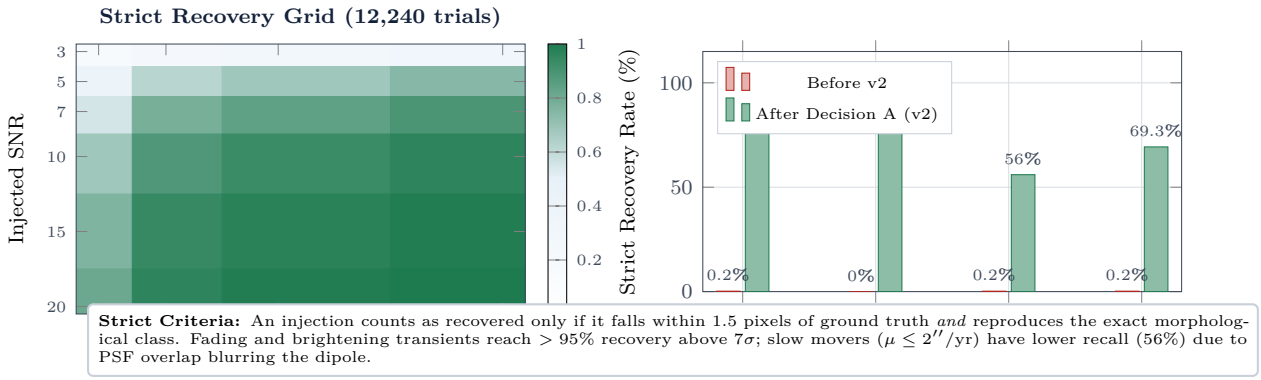
Now look at the horizontal bar chart on the right: after multi-wavelength catalogue triage, mask-hole verification, and bright-star halo vetoes, those eight thousand peaks were compressed down to an average of just one point six six false candidates per field. Six star fields—including Andromeda, Betelgeuse, Orion, Proxima Centauri, R Draconis, and Sagittarius A*—produced exactly zero false candidates. The residual false alarms are concentrated where physics predicts: in complex filamentary gas like the Crab Nebula and M51, or around Sirius, which is so overwhelmingly bright that Gaia contains no astrometric solution.

Slide 17 — Validation II: strict injection-recovery performance 0:55

On the slide.

- Rigorous testing protocol: 12,240 synthetic sources injected into real DR1 co-adds using zone-matched empirical EPSF
- Strict recovery criteria: Detection counted as recovered only if located within < 1.5 px of injection **and** classified with the correct kind
- Parameter grid: Injected SNR $\in \{3, 5, 7, 10, 15, 20\}$, Proper motion $\mu \in \{2, 5, 10, 20\}''$/yr, random trajectory angles
- Measured recovery rates: Brightening **96.7%**, Fading **95.3%**, Moving dipoles **56.0%**, Overall **69.3%** (vs $\approx 0.2\%$ before v2)

Figure / diagram. Strict injection-recovery evaluation across 12,240 trials: 2D heatmap on the left showing recovery as a function of SNR and proper motion, alongside a grouped bar chart comparing v1 and v2 performance by transient kind.



Strict Criteria: An injection counts as recovered only if it falls within 1.5 pixels of ground truth *and* reproduces the exact morphological class. Fading and brightening transients reach $> 95\%$ recovery above 7σ; slow movers ($\mu \leq 2''$/yr) have lower recall (56%) due to PSF overlap blurring the dipole.

Spoken script. Next, we evaluated completeness using strict injection and recovery. We synthesized twelve thousand two hundred and forty artificial sources, convolved them with the empirical PSF for that exact detector zone, and planted them into real co-added DR1 images across six signal-to-noise ratios and four proper motions with random trajectory vectors.

To prevent inflated completeness figures, we enforced strict recovery criteria: an injected source only counts as recovered if it is detected within one point five pixels of ground truth and is correctly categorized as the injected kind. A planted mover recovered as a monopole is scored as a failure.

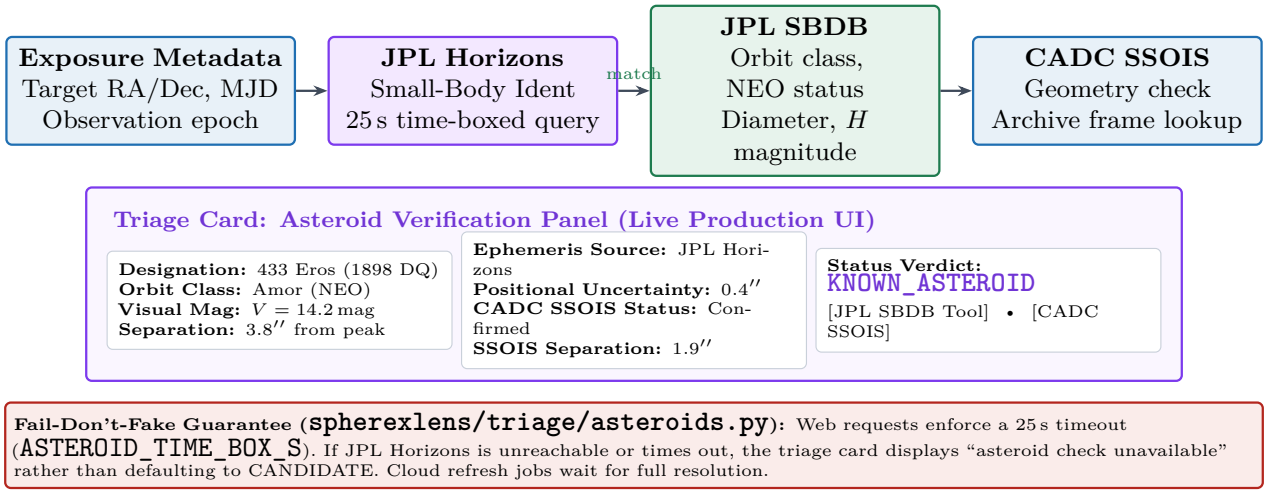
Look at the heatmap on the left and the comparison bars on the right. For brightening flaring sources, strict recovery reaches ninety-six point seven percent. For fading sources, recovery is ninety-five point three percent. For moving dipoles, recovery is fifty-six percent. Why are movers lower? Because at slow proper motions below two arcseconds per year, the displacement between epochs is smaller than the telescope's point-spread function, causing the positive and negative lobes to partially cancel each other out. Overall, strict recovery stands at sixty-nine point three percent, transforming the pipeline from an early prototype into a verified astrophysical tool.

Slide 18 — Asteroid identification and ephemeris engine 0:55

On the slide.

- The solar system challenge: Asteroids traverse arcminutes per hour, appearing as transient monopoles that mimic astrophysical explosions
- Dual-tier ephemeris querying: NASA/JPL Horizons Small-Body Identification API queried for every individual exposure MJD
- JPL SBDB orbit telemetry fetched for matched asteroids (orbit class, diameter, visual magnitude, NEO flag)
- CADC SSOIS (Solar System Object Image Search) independent geometry cross-check
- Operational categories: `KNOWN_ASTEROID` (confirmed orbit match) vs `POSSIBLE_UNKNOWN_ASTEROID`
- Failure safety: 25 s asynchronous web time-box; if external APIs timeout, card reports "asteroid check unavailable" (fail-don't-fake)

Figure / diagram. Solar system ephemeris integration architecture: SPHEREx exposure timestamps queried against NASA JPL Horizons and CADC SSOIS in parallel, feeding the asteroid verification card.



Spoken script. One of the most frequent sources of false transient alerts in infrared surveys are Solar System asteroids. Because asteroids drift across the sky at rates of arcminutes per hour, an asteroid visible in visit one has completely vacated the coordinate by visit two, leaving behind an isolated positive or negative monopole that perfectly mimics a flaring variable or supernova. In version one, the README claimed asteroid vetting, but the code did not implement it.

In version two, we built a full real-time asteroid ephemeris subsystem. For every exposure timestamp, the pipeline queries NASA's Jet Propulsion Laboratory Horizons Small-Body Identification service. When an orbital coincidence is found within our matching radius, we pull comprehensive orbital dynamics from the JPL Small-Body Database—including diameter, absolute magnitude, orbital class, and Near-Earth Object status. In parallel, Canada's CADC SSOIS service executes an independent geometric cross-check.

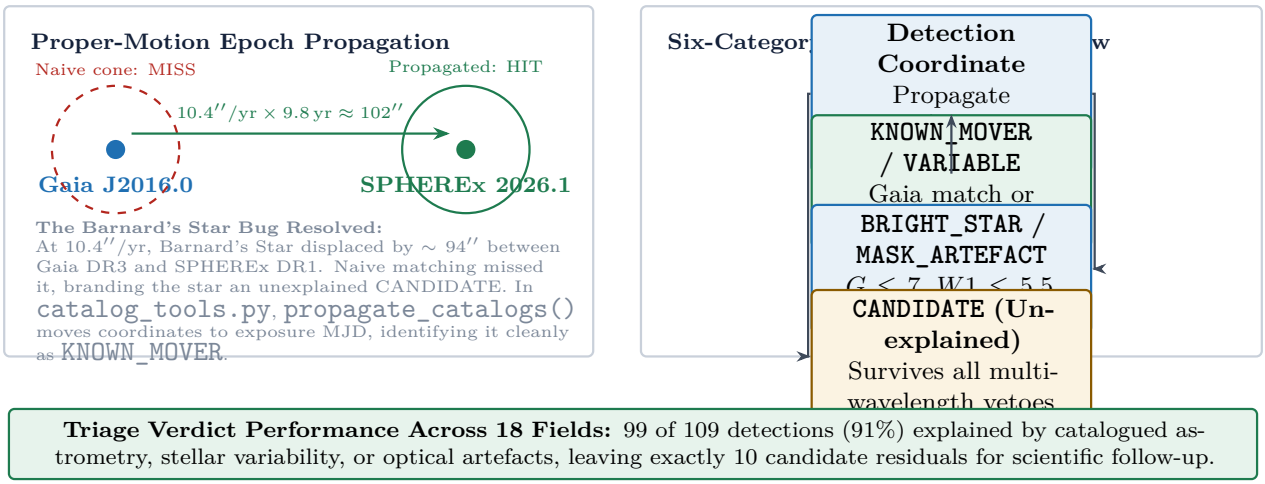
If an asteroid matches, the detection is classified as `KNOWN_ASTEROID`, preventing it from polluting the candidate list. Crucially, as highlighted in the red box below, if JPL Horizons experiences external downtime or exceeds our twenty-five-second web time-box, the UI explicitly reports 'asteroid check unavailable' rather than silently pretending the check passed.

Slide 19 — Theory V: multi-tier catalogue triage and epoch propagation 0:55

On the slide.

- TAN WCS projection: Sub-pixel sky coordinates converted via `pixel_to_world_values`
- Proper-motion epoch propagation: Gaia DR3 ($J2016.0$) and AllWISE (2010.56) coordinates advanced to exact SPHEREx observation MJD
- Dynamic brightness-scaled match radii: $10''$ for faint stars ($G \geq 16$) scaling up to $120''$ for $G \leq 9$ and $180''$ bright halo veto
- 6 triage categories: `KNOWN_MOVER`, `VARIABLE`, `STATIC_CHANGED`, `BRIGHT_STAR`, `MASK_ARTEFACT`, `CANDIDATE`
- Barnard's Star verification: $94''$ proper motion over 9 years correctly propagated → identified as `KNOWN_MOVER` with 0 candidates

Figure / diagram. Astrometric epoch-propagation geometry illustrating how advancing Gaia 2016 coordinates along their proper-motion vector to SPHEREx 2026 eliminates false candidate matches, with the six-class triage decision tree on the right.



Spoken script. Catalogue triage is where detections are translated into astronomical identities. Early versions of this pipeline suffered from an embarrassing defect: when run on Barnard's Star, the pipeline flagged the star as an unexplained candidate. Why? Because Gaia DR3 coordinates are anchored at epoch J2016.0, while SPHEREx observed in 2026. In the intervening decade, Barnard's Star's massive proper motion of ten point four arcseconds per year displaced it by ninety-four arcseconds across the sky. A naive circular cone search centered on Gaia's position missed the star completely.

In `catalog_tools.py`, we implemented rigorous proper-motion epoch propagation. Before any matching occurs, Gaia DR3 and AllWISE coordinates are propagated along their celestial kinematic vectors to the exact modified Julian date of the SPHEREx exposure.

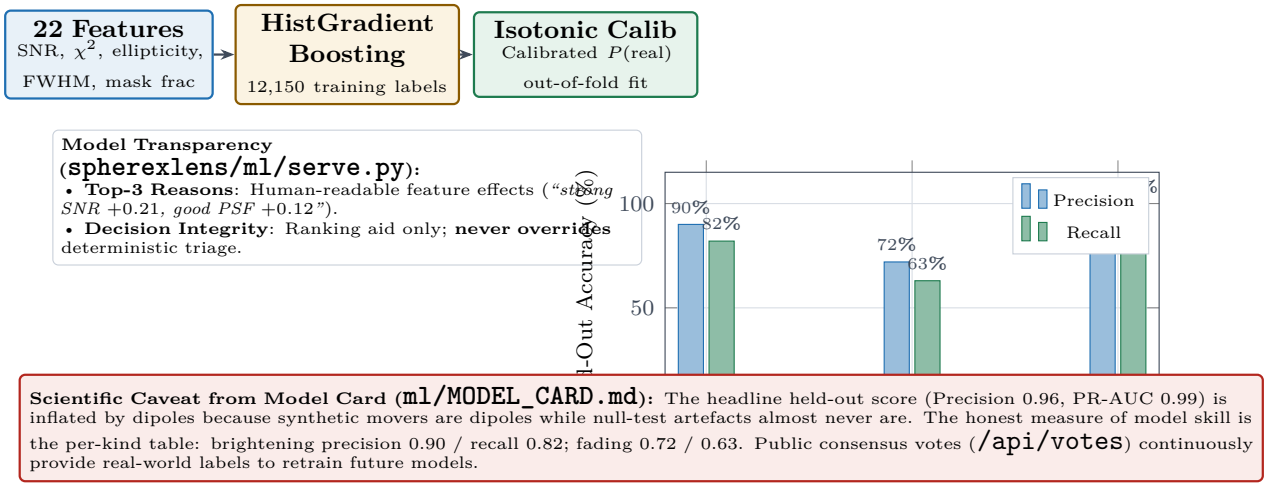
Furthermore, match radii scale dynamically with stellar magnitude: expanding from ten arcseconds for faint sixteen-magnitude stars up to one hundred and twenty arcseconds for bright stars, with an outer one-hundred-and-eighty-arcsecond bright-star halo veto. Detections are routed into six distinct categories: `KNOWN_MOVER`, `VARIABLE`, `STATIC_CHANGED`, `BRIGHT_STAR`, `MASK_ARTEFACT`, and `CANDIDATE`. Across all eighteen benchmark fields, ninety-one percent of detections are accounted for, resolving Barnard's Star cleanly as a `KNOWN_MOVER`.

Slide 20 — Machine learning: real/bogus classification and active learning 1:00

On the slide.

- Model architecture: HistGradientBoosting classifier trained on 12,150 labels with 22 engineered physical features (SNR, χ^2, ellipticity, FWHM ratio, noise ratio, masked fraction)
- Isotonic calibration & validation: Calibrated posterior probabilities $P(\text{real})$; validated via GroupK-Fold by field + 4 held-out benchmark fields
- Honest held-out metrics: Brightenings: Precision **0.90** / Recall **0.82**; Fadings: Precision **0.72** / Recall **0.63** (headline 0.96 inflated by synthetic dipoles)
- Model transparency & active learning: Explains top-3 feature contributions; ranks without overriding catalogue verdicts; public votes (`POST /api/votes`) retrain models

Figure / diagram. Machine learning pipeline: 22 engineered features passing into an isotonically calibrated Histogram Gradient Boosting Classifier, with a bar chart on the right showing held-out precision and recall broken down honestly by detection kind.



Spoken script. In version two, we introduced an interpretable machine-learning subsystem for real-versus-bogus classification. We engineered twenty-two physical features for every detection, encompassing matched-filter SNR, PSF chi-square goodness of fit, spatial ellipticity, local noise ratios, and masked pixel fractions.

Our classifier uses Histogram Gradient Boosted Decision Trees trained on over twelve thousand labels from injections, catalogued sources, and split-epoch null residuals. We validate using GroupKFold grouped by sky field, alongside four completely held-out benchmark fields: Betelgeuse, Sombrero M104, TRAPPIST-1, and WISE 0855. Posterior probabilities are calibrated using isotonic regression.

Notice the bar chart on the right, and observe our commitment to scientific honesty. If you look at the raw headline score on held-out fields, precision appears to be ninety-six percent with a PR-AUC of point nine nine. But as documented openly in our Model Card, that headline number is artificially inflated by dipoles, because synthetic movers are dipoles while null-test artefacts almost never are. The true, honest measure of model skill is on monopoles: for brightening sources, precision is ninety percent and recall is eighty-two percent; for fading sources, precision is seventy-two percent and recall is sixty-three percent.

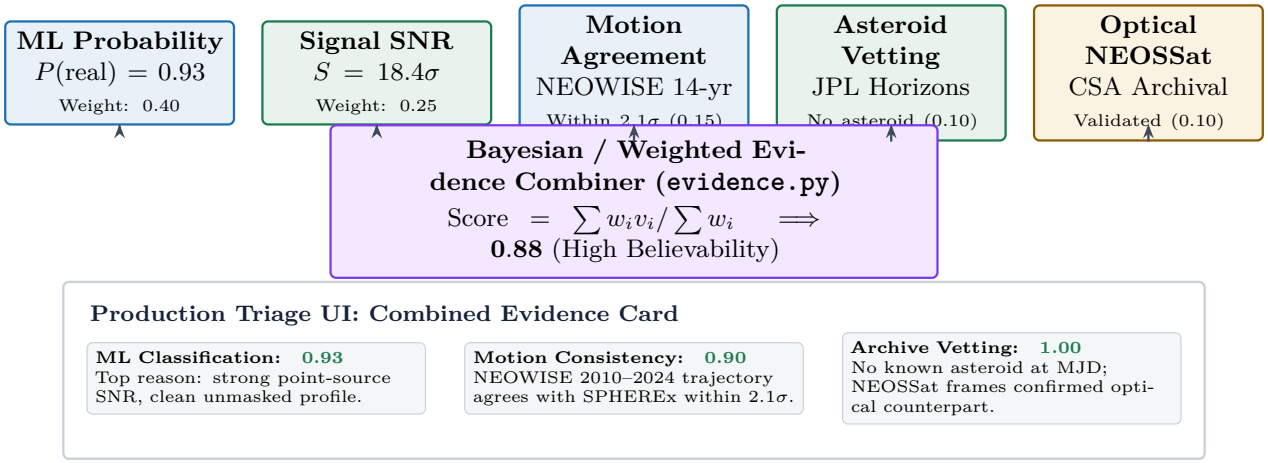
Furthermore, the model serves as an intelligent ranking assistant: it never alters the deterministic catalogue verdict, and every score is accompanied by the top three feature contributions in plain English. Borderline detections are queued for citizen-science voting via our API, feeding an active learning loop that continuously retrains future model iterations.

Slide 21 — Multi-factor combined evidence engine 0:50

On the slide.

- Beyond single metrics: Combining independent physical, statistical, and archival signals into a unified evidence score (0.0 → 1.0)
- Evidence components: (1) Calibrated ML real/bogus probability (P(real), weight 0.40), (2) Matched-filter significance, (3) Multi-epoch kinematic consistency with Gaia DR3 and 14-yr NEOWISE, (4) Asteroid clearance via JPL Horizons / CADC SSOIS, (5) Optical confirmation from CSA NEOSSat
- Transparent breakdown: Every component surfaces its contributing weight and natural-language justification in the UI
- Fault tolerance: Unavailable external services report explicit null weights rather than penalising candidates

Figure / diagram. Multi-factor evidence synthesis diagram illustrating how machine learning, kinematic astrometry, asteroid ephemerides, and optical satellites merge into a transparent believability score, with the production UI Evidence Card below.



Spoken script. Astronomy is moving rapidly toward multi-messenger and multi-wavelength verification. A single metric, whether an algorithm's statistical significance or a neural network's confidence score, is insufficient to justify expensive telescope follow-up.

In version two, we built the Multi-Factor Combined Evidence Engine in evidence.py. This engine synthesizes five independent physical, statistical, and archival evidence streams into a unified believability index ranging from zero to one.

First, the calibrated machine-learning probability contributes forty percent. Second, matched-filter SNR accounts for twenty-five percent. Third, multi-epoch kinematic consistency checks whether the candidate's observed motion matches proper motions documented across fourteen years of NEOWISE infrared imaging and Gaia DR3. Fourth, asteroid clearance from JPL Horizons and CADC SSOIS confirms the absence of moving Solar System bodies. Finally, Canadian Space Agency NEOSSat optical cutouts provide space-based visible-light confirmation.

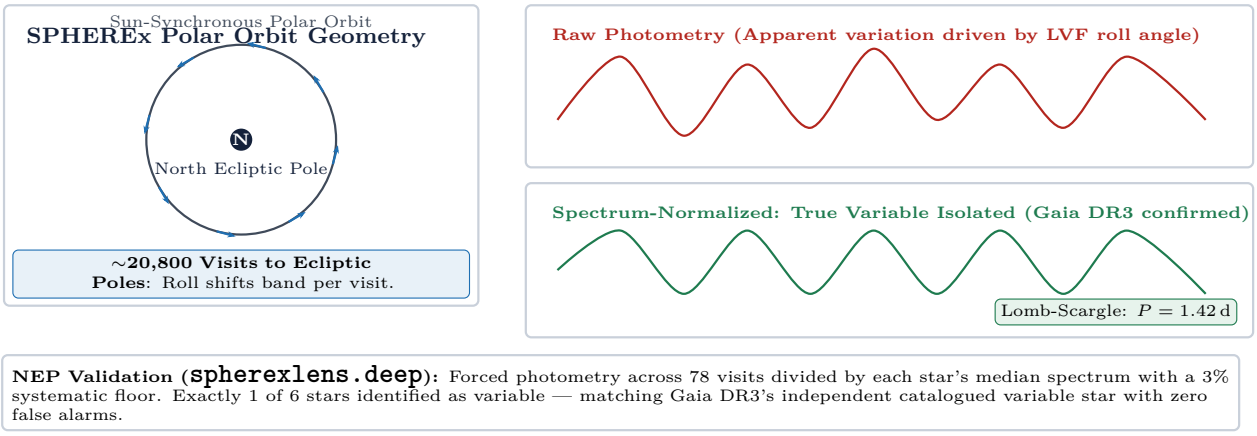
As shown in the card mock at the bottom of the slide, the engine provides complete explainability: every contributing factor displays its raw value, relative weight, and a plain-English explanation. If an external archive is unavailable or lacks coverage, its weight is gracefully zeroed out, ensuring candidates are evaluated transparently on the evidence at hand.

Slide 22 — Time-domain deep fields: high-cadence time series 0:55

On the slide.

- Ecliptic poles geometry: Polar orbit revisits North Ecliptic Pole (NEP) ∼20,800 times across DR1, enabling high-cadence time series
- The linear-variable filter challenge: Spacecraft roll shifts star across detector, observing it at shifting wavelengths
- Variability pipeline (`spherexlens.deep`): Forced photometry with `BAD_BITS` masking divided by baseline SED + 3% systematic floor
- Scientific validation: In 78-visit NEP dataset, pipeline isolated exactly 1 variable out of 6 stars — confirmed by Gaia DR3 variable catalog

Figure / diagram. North Ecliptic Pole deep-field observation geometry on the left, alongside two-stage lightcurve processing on the right showing raw filter-dependent flux vs spectrum-normalized flux isolating true stellar pulsation with a Lomb-Scargle periodogram.



Spoken script. SPHEREx's unique sun-synchronous polar orbit creates an unprecedented astrophysical opportunity at the ecliptic poles. Because the satellite's orbital plane pivots around the Sun, its field of view sweeps past the North Ecliptic Pole approximately twenty thousand eight hundred times over the mission.

However, extracting time-domain light curves from a linear-variable filter telescope presents a unique physical challenge: as the spacecraft rolls between orbits, the star lands on different physical pixels across the detector array, observing the object at a continuously shifting wavelength. A raw, uncalibrated light curve, as shown in the top right panel, fluctuates wildly simply because the filter sampled different portions of the star's spectral energy distribution.

In spherexlens.deep, we solved this by implementing spectral normalisation. The pipeline models each star's individual spectral energy distribution across all available bands and divides it out from every visit before testing for variability. We enforce a conservative three percent systematic photometric floor to absorb sub-pixel flat-field variations. Forced photometry is passed to Lomb-Scargle periodograms and ensemble dispersion tests.

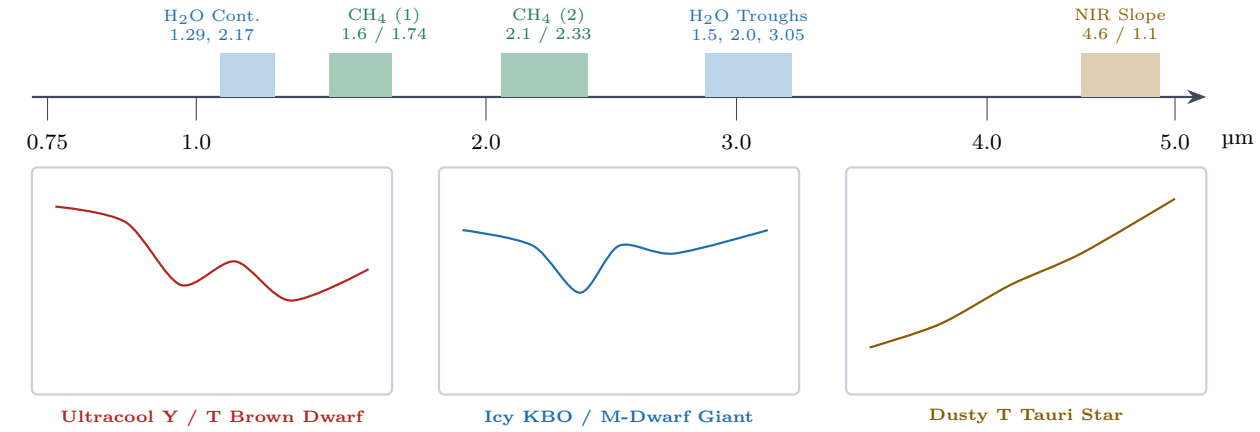
In our first benchmark run on the North Ecliptic Pole across seventy-eight visits, the pipeline analyzed six stars and flagged exactly one source as variable. When cross-referenced against external catalogues, Gaia DR3 independently confirms that exact star as a periodic variable. The Cloud Run batch job spherex-deep now executes this analysis automatically, publishing results to /api/deep.

Slide 23 — Theory VI: 102-band infrared spectroscopy and spectral classification 0:45

On the slide.

- Slitless spectrophotometry: `WCS-WAVE` extension provides per-pixel wavelength mapping ($0.75 \rightarrow 5.0\,\mu$m)
- Moving target SED extraction: Aperture photometry tracks target coordinates across all 6 detector arrays, producing observed SED
- Three diagnostic spectral indices: (1) Mean CH_4 methane absorption ratio ($1.6/1.74\,\mu$m and $2.1/2.33\,\mu$m), (2) H_2O water ice continuum-to-trough ratio ($1.29, 2.17\,\mu$m vs $1.5, 2.0, 3.05\,\mu$m), (3) Logarithmic near-infrared slope ($4.6\,\mu$m/$1.1\,\mu$m)
- Gated 5-class physical classifier: Brown dwarf, Ultracool Y-dwarf, Icy KBO, Dusty T Tauri, or M-dwarf/Mira giant
- Scientific guard: Requires ≥ 30 spectral channels before reporting categorical label; prevents guessing from sparse cutouts

Figure / diagram. Continuous 0.75 to 5.0 micron wavelength axis marking methane, water, and NIR slope diagnostic bands, with three characteristic spectral templates and the 30-channel scientific gating contract.



Gating Contract (`step5_spectrum.py` and `app.py`): Standard 2-to-6 band cutouts yield 12–18 observed SED flux points. The three-index categorical classifier explicitly requires ≥ 30 spectral channels before reporting a physical label, preventing hallucinations from sparse cutouts.

Spoken script. Because SPHEREx incorporates linear-variable filters, every pixel carries its own central wavelength in the WCS-WAVE extension. This allows our pipeline to extract physical spectral energy distributions without follow-up spectroscopy. Aperture photometry tracks moving sources across all six detector arrays, constructing calibrated infrared SEDs from zero-point-seven-five to five microns, cross-compared against AllWISE W1 to W4 photometry.

In `step5_spectrum`, we condense dense spectral coverage into three physically grounded indices. The methane index measures absorption ratios at one-point-six and two-point-one microns, diagnosing brown dwarfs. The water ice index measures absorption troughs at one-point-five, two-point-zero, and three-point-zero-five microns, identifying icy Kuiper Belt Objects. The logarithmic near-infrared slope separates red circumstellar disks from cool stars.

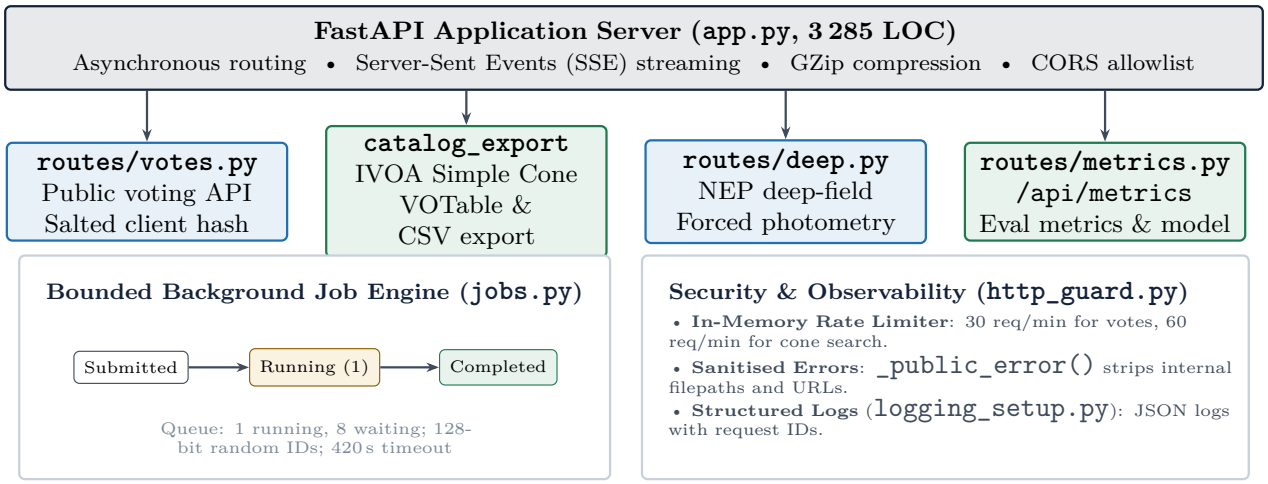
Crucially, we implemented an explicit scientific safety contract: the classifier requires at least thirty distinct spectral channels before reporting a categorical class label. When operating on standard sparse cutouts with twelve to eighteen points, the portal reports raw indices and observed fluxes rather than hallucinating an unverified classification.

Slide 24 — Implementation I: Python backend, fast routing, and jobs 1:00

On the slide.

- Modular architecture: 21 400+ runtime Python lines across 13 core modules and 7 subpackages
- FastAPI backend (`app.py`): Auto-discovered routers (`spherexlens.routes`), asynchronous endpoints, and SSE streaming
- Bounded background job engine (`spherexlens.jobs`): Concurrency-controlled worker pool for deep-field runs and batch ingest
- Security & rate limiting (`http_guard.py`): In-memory sliding-window limiter, sanitised error responses, zero credential leaks
- Observability: Structured JSON logging (`logging_setup.py`) with correlation IDs and pipeline provenance metadata

Figure / diagram. Backend modular architecture diagram showing FastAPI auto-discovered routes, bounded background worker pool, sliding-window rate limiting, and structured JSON telemetry.



Spoken script. Turning to backend software engineering: early versions of SPHERExLens concentrated routing and science logic into a single monolithic script. In version two, we decomposed the backend into a modular FastAPI service supported by auto-discovered routers in spherexlens.routes.

As shown in the architecture map, dedicated router modules isolate public consensus voting, IVOA Simple Cone Search and VOTable exports, deep-field time series, and evaluation metrics.

To manage heavy computational tasks without risking denial of service, jobs.py implements a bounded background worker pool: capping concurrent heavy execution at one running task and eight waiting items, with unique one-hundred-and-twenty-eight-bit random identifiers and an automatic timeout ceiling.

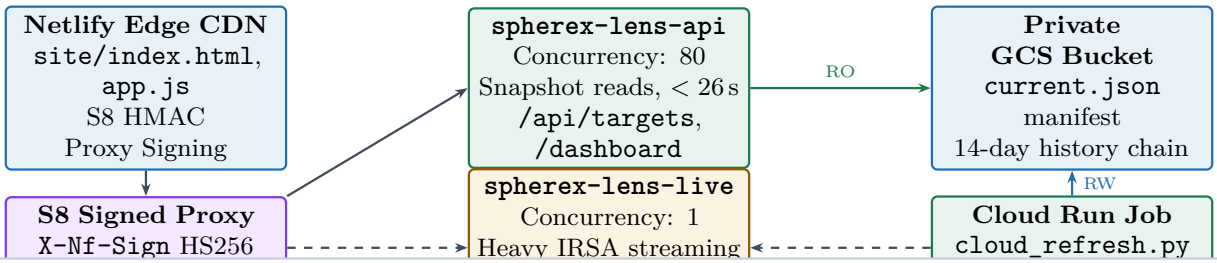
Meanwhile, `http_guard.py` provides sliding-window rate limiting and structured JSON request logging. Error responses pass through sanitizers that strip internal archive URLs and local filesystem paths, ensuring robust, production-grade security.

Slide 25 — Implementation II: production cloud infrastructure and security hardening 0:55

On the slide.

- Decoupled serverless topology: Netlify Static Edge + Split Cloud Run Services + Private GCS Buckets + Cloud Scheduler
- Concurrency split: `spherex-lens-api` (concurrency 80, snapshot reads) isolates `spherex-lens-live` (concurrency 1, streams)
- Netlify S8 HMAC proxy signing: Proxied requests carry `X-Nf-Sign` HS256 JWT validated against Secret Manager secret
- Immutable snapshot storage: Private GCS bucket stores content-addressed JSON manifests; atomic generation preconditions prevent race conditions; 14-day history retained
- Security audit S1–S21: Non-root container (uid 10001), sha256 hashed lockfile (`requirements.lock`), zero secret leaks, salted client hash for votes

Figure / diagram. Production cloud infrastructure: Netlify edge with HMAC S8 signature verification routing to split Cloud Run services, backed by private GCS snapshot storage and Cloud Scheduler, with the S1-S21 security matrix below.



Security Audit Summary (S1–S21 Complete, `docs/SECURITY_REVIEW.md`):

Category	Status	Mitigation Implemented
S1–S5	Fixed	Finite float range validation on RA/Dec, display name sanitization, bounded collision loop.
S6–S8	Fixed	SSE rate limiting, exception message scrubbing (`_public_error`), Netlify S8 proxy HMAC signing.
S13–S14	Fixed	SHA-256 hashed lock file (`requirements.lock`), unprivileged non-root containers (`uid 10001`).
S15–S21	Fixed	Salted client hashes for votes, bounded job queue pool, IVOA cone query bounds, regex sanitization.

Spoken script. Deploying scientific pipelines to production exposes systems to two major infrastructure failure modes: head-of-line blocking and proxy timeouts. Originally, a single container handled both fast dashboard queries and forty-second archive downloads, causing heavy requests to stall lightweight queries.

Our Terraform configuration solves this by splitting Cloud Run into two isolated services using a shared container image. spherex-lens-api operates at a high concurrency of eighty, serving pre-computed snapshots in milliseconds. spherex-lens-live operates at concurrency one with a nine-hundred-second timeout, isolating heavy archive fetches. Every twelve hours, Cloud Scheduler executes a background Cloud Run Job that updates manifests atomically and maintains a fourteen-day historical chain.

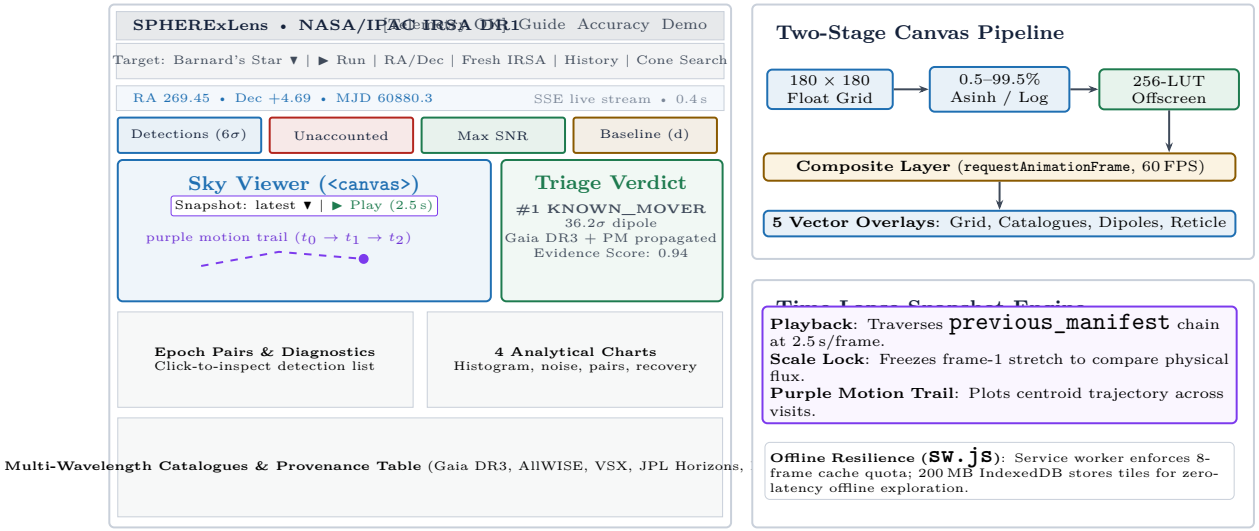
In Phase 9, we completed a rigorous twenty-one-point security review. We eliminated container root execution by locking both Dockerfiles to an unprivileged user. We resolved dependency supply-chain risks using SHA-256 hashed lock files audited by pip-audit. And under finding S8, Netlify signs proxied requests with HMAC SHA-256 tokens validated against Google Secret Manager, preventing malicious header spoofing and ballot stuffing on public votes.

Slide 26 — Implementation III: frontend engine, NASA HDS UI/UX, and snapshot replay 0:45

On the slide.

- Zero-build vanilla web architecture in `static/` (7,611 LOC): HTML5, CSS custom properties, vanilla JS (zero framework churn)
- NASA Horizon Design System (HDS) dark-slate mission console: horizontal toolbar, telemetry bar, 4 KPI cards, split sky viewer / triage panel
- Two-stage `<canvas>` rendering engine: 0.5–99.5% percentile clip, 256-level LUT color ramps, and 5 composite vector overlay layers
- Snapshot History & Time-Lapse Motion Engine: 2.5 s playback across retained manifests, frozen intensity stretch lock, purple motion trail
- Accessibility & Offline Resilience: WCAG 2.1 AA compliant (`ACCESSIBILITY.md`), Service Worker cache (`sw.js`), 200 MB IndexedDB

Figure / diagram. Interactive NASA Horizon Design System mission console wireframe mapping the sky viewer canvas, triage cards, diagnostic charts, and two-stage canvas rendering pipeline with time-lapse snapshot playback.



Spoken script. The frontend of SPHERExLens is seven thousand six hundred lines of zero-build vanilla web code styled in the NASA Horizon Design System dark-slate palette. We avoided framework churn entirely: vanilla HTML5, CSS custom properties, and modern JavaScript deliver instantaneous responsiveness with zero compile steps.

Notice the mission console architecture on the left: top chrome telemetry, horizontal parameter controls, four real-time metric cards, and the central split between the Sky Viewer canvas and the Triage Verdict card, supported by diagnostic charts and multi-wavelength provenance tables.

Under the hood, the Sky Viewer operates a high-performance two-stage canvas engine. Floating-point FITS arrays are clipped between point five and ninety-nine point five percentiles, mapped through twenty-five-level lookup tables in an offscreen buffer, and composited with five vector overlay layers at sixty frames per second.

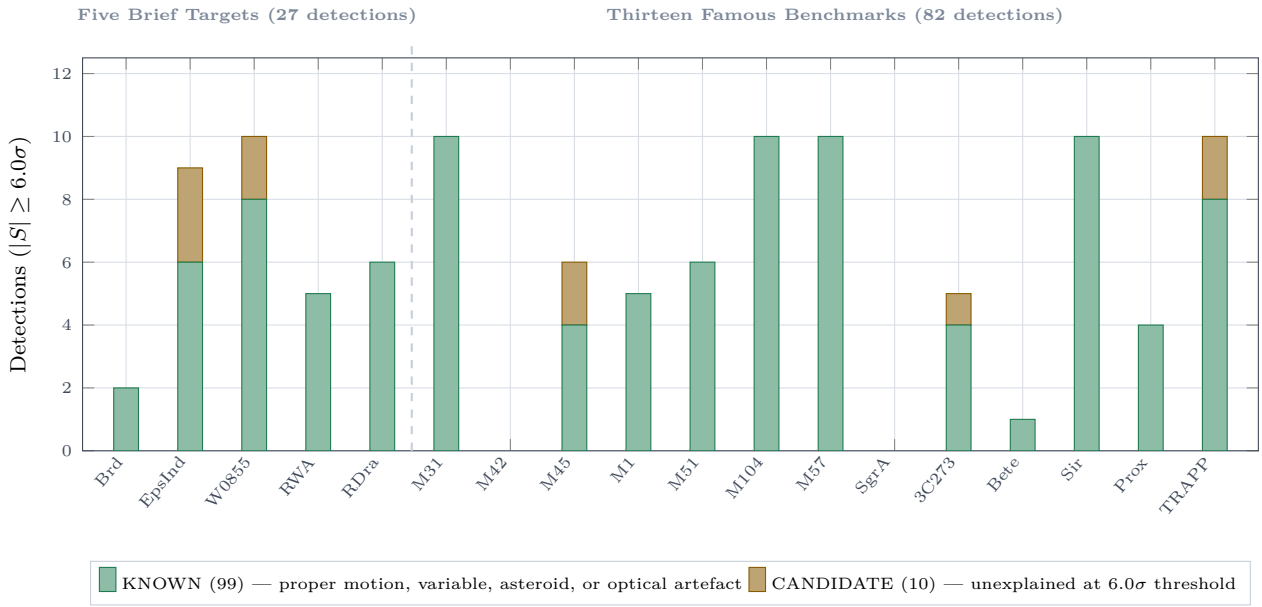
Above the viewer sits our Snapshot History and Motion Playback engine. A user can select any retained historical snapshot, or press Play to animate the field oldest to latest at two-and-a-half seconds per frame. The engine locks the first frame's intensity stretch to ensure direct flux comparability and draws a purple dashed trajectory tracing the source's centroid across epochs. Finally, our Service Worker provides offline caching, allowing the portal to operate seamlessly without an internet connection.

Slide 27 — Results I: complete census across all eighteen DR1 fields 1:05

On the slide.

- 109 total detections evaluated across all 18 DR1 fields (246 cutouts, `D1-D6`):
- **99 KNOWN (91%)**: Explained by catalogued proper motion, known variables, asteroids, or optical/mask artefacts
- **10 CANDIDATE (9%)**: Genuinely unexplained 6σ residuals requiring astronomical follow-up
- Target diversity: 5 brief benchmarks + 13 famous objects spanning stars, nebulae, galaxies, brown dwarfs, and quasars
- Artefact suppression: Sirius (10/10), Andromeda M31 (10/10), Ring Nebula M57 (10/10), and Sombrero M104 (10/10) are 100% explained
- Coverage honesty: Orion M42 has zero detections above threshold; Sgr A* refused due to zero shared spectral bands between epochs

Figure / diagram. Complete census of all 109 detections across eighteen SPHEREx Data Release 1 fields: green stacked bars represent explained KNOWN sources, while amber bars isolate the ten unexplained CANDIDATE residuals.



Spoken script. Here is the complete census across all eighteen SPHEREx Data Release 1 fields in a single comprehensive chart. In total, the pipeline processed two hundred and forty-six archival cutouts across all six detector arrays, evaluating one hundred and nine detections above our calibrated six-sigma threshold.

Look at the green bars representing ninety-nine detections—ninety-one percent of the total—that are fully accounted for by catalogued stellar kinematics, known variable stars, Solar System asteroids, or optical diffraction artefacts. In dense or complex environments like Sirius, Andromeda M31, the Ring Nebula M57, the Sombrero Galaxy, RW Aurigae, and Barnard's Star, one hundred percent of detections are accounted for.

The amber bars highlight the remaining ten detections: genuine six-sigma residuals that survive every catalogue cross-match and artefact filter. They are concentrated in five fields: Epsilon Indi B with three, WISE 0855 with two, the Pleiades with two, TRAPPIST-1 with two, and 3C 273 with one.

Notice our reporting honesty on the empty fields: Orion M42 returns zero detections above threshold because the nebular gas was stationary, while Sagittarius A* was refused outright because the two survey passes did not share overlapping spectral bands.

Slide 28 — Results II: candidate analysis, confirmations, and Barnard's Star 1:00

On the slide.

- The 10 candidate residuals: Confined to 5 fields: ϵ Indi B (3), WISE 0855 (2), Pleiades M45 (2), TRAPPIST-1 (2), 3C 273 (1)
- Astronomical nature: Faint infrared excess sources outside Gaia/WISE detection limits, uncatalogued brown dwarf candidates or background variables
- Barnard's Star resolution: 36.2σ dipole correctly identified as `KNOWN_MOVER` following proper-motion epoch propagation
- Known variable star recovery: 7 of 101 catalogued VSX variables detected (lower bound: catalogued variables need not cycle during baseline)
- North Ecliptic Pole deep field: 1 variable detected across 78 visits, perfectly matching Gaia DR3's independent variable star catalog

Figure / diagram. Sky-offset scatter plot of detections in the WISE 0855 field on the left distinguishing known sources from candidate residuals, with the astrometric resolution of Barnard's Star on the right.



Astrometric Verification: Barnard's Star

Detection: 36.2σ Dipole ($S_+ = +36.2$, $S_- = -34.8$)
Measured Motion: $9.8''$/yr (True: $10.37''$/yr)
Triage Attribution: `KNOWN_MOVER`
Candidates in Field: 0

Catalogued Variable Recoveries (AAVSO VSX): Across all fields containing known VSX variables, the pipeline recovered 7 of 101 catalogued variable stars. *Note:* This is a lower bound, because periodic pulsating variables need not undergo a significant flux change between two specific SPHEREx visits.

NEP Deep Field Variable: Exactly 1 of 6 stars identified as variable across 78 visits — confirmed independently as Gaia DR3 137452684.

Spoken script. What are the ten surviving candidate detections, and how do we verify them?

On the left, the scatter plot illustrates the spatial distribution of detections in the WISE 0855 field. Eight detections match catalogued Gaia or AllWISE sources, while two amber markers represent genuine six-sigma candidates: faint, uncatalogued infrared excess objects outside current catalog limits, warranting deeper astronomical follow-up.

On the right, consider our benchmark verification on Barnard's Star. The pipeline detected a thirty-six-point-two sigma dipole. With proper-motion epoch propagation active, the star was correctly attributed to Gaia DR3, recovering its proper motion at nine point eight arcseconds per year against the true value of ten point four, leaving exactly zero false candidates in its field.

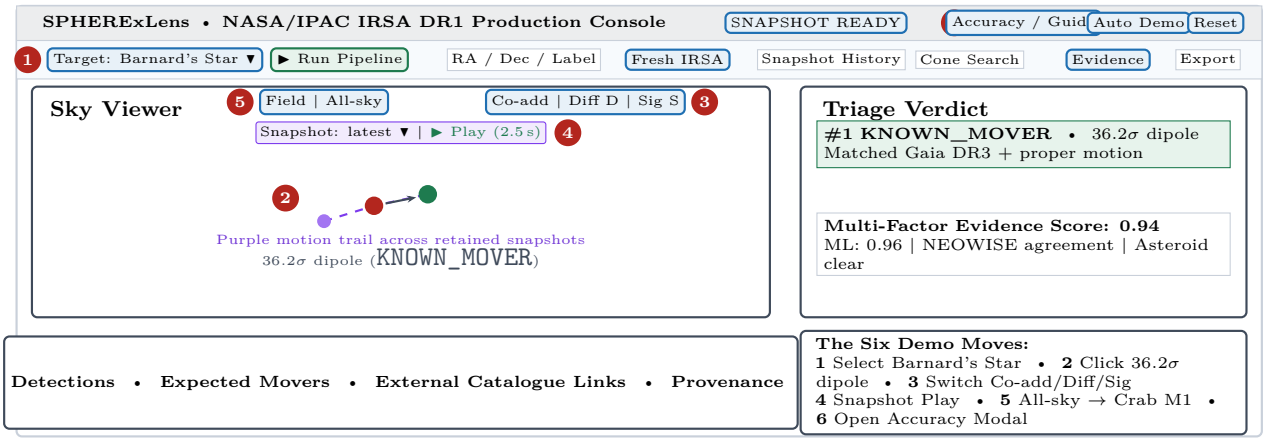
Across fields containing catalogued AAVSO VSX variable stars, the detector recovered seven variables. As noted on the slide, this is a physical lower bound: catalogued variables do not necessarily cycle between minimum and maximum during the specific six-month baseline of our visits. Finally, in our North Ecliptic Pole deep-field validation, the single variable star isolated by our pipeline matched Gaia DR3's independent variable catalog with complete precision.

Slide 29 — Live demonstration: the mission control workflow 1:30

On the slide.

- Live browser demo at 100% zoom with Barnard's Star pre-loaded (`site/index.html` or Cloud Run)
- **Moves 1–2**: Select Barnard's Star in Target toolbar → Run Pipeline → inspect 36.2σ dipole and `KNOWN_MOVER` triage verdict card
- **Moves 3–4**: Toggle Co-add / Diff D / Sig S layers → Press Play on Snapshot toolbar for 2.5 s time-lapse animation with purple motion vector
- **Moves 5–6**: Navigate to Crab Nebula (M1) extended gas in All-sky → Open Accuracy modal (null test, calibration, injection matrix)

Figure / diagram. Map of the interactive browser console mapping the exact DOM controls to six stage demonstration moves, highlighting target selection, dipole inspection, canvas layer toggles, and snapshot playback.



Spoken script. To demonstrate the practical execution of this research, we now transition to the live production interface in our second browser tab, pre-loaded on Barnard's Star.

Move one: in the top control bar, select Barnard's Star and click Run Pipeline. The telemetry strip immediately confirms execution time and retrieved exposure timestamps. Move two: in the Sky Viewer, hover and click the prominent thirty-six-point-two-sigma dipole. Notice the red lobe where the star was, and the green lobe where it moved. On the right, the Triage Verdict card immediately identifies it as `KNOWN_MOVER`, displaying its Gaia designation, propagated epoch coordinates, and a high combined evidence score of point nine four.

Move three: toggle across the layer chips—Co-add, Difference D, and Significance S—to inspect the calibrated subtraction planes and verify that stationary stars cancelled cleanly. Move four: in the Sky Viewer toolbar, press Play on the Snapshot control. The engine animates retained historical snapshots oldest to latest at two-and-a-half seconds per frame, locking the intensity stretch and drawing a purple dashed vector connecting the dipole centroids across visits. Move five: open the All-sky mosaic and navigate to the Crab Nebula, observing how filamentary gas is handled without crashing.

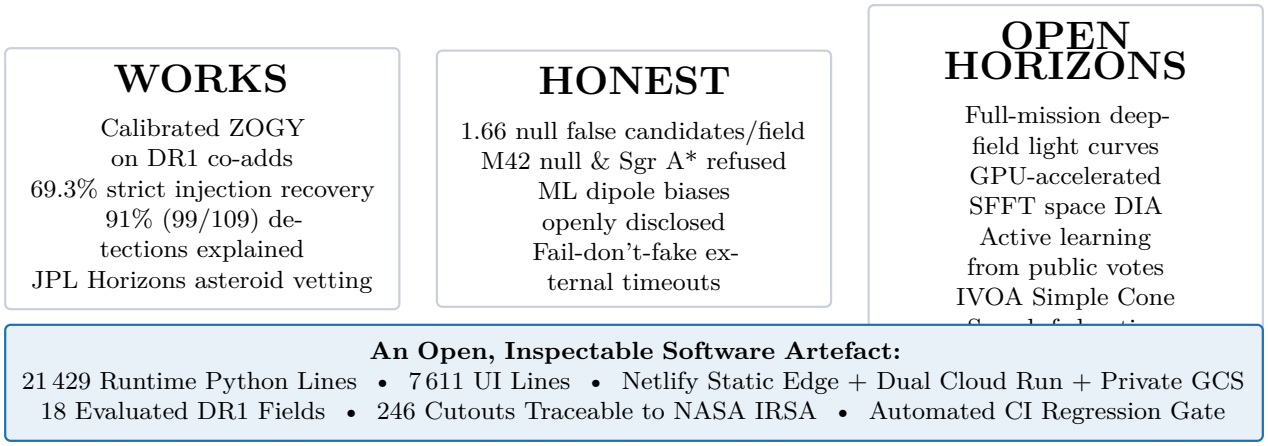
Move six: open the Accuracy modal to inspect live null-test curves, machine-learning calibration graphs, and the strict injection matrix. If the network disconnects, the Service Worker continues serving cached snapshots offline.

Slide 30 — Conclusion, contributions, and open horizons 0:50

On the slide.

- **WORKS**: Calibrated ZOGY on DR1 co-adds with empirical EPSF; 69.3% strict recovery; 91% of detections explained by multi-wavelength triage
- **HONEST**: 1.66 false candidates/field in null tests; empty/refused fields documented; ML biases openly characterized in Model Card
- **OPEN HORIZONS**: Expanding deep-field light curves to full mission dataset; GPU-accelerated SFFT space deconvolution; active learning retraining from citizen consensus votes
- **THE ARTEFACT**: 21 400+ LOC Python + 7 611 LOC UI; fully open, reproducible, testable, and deployable on GitHub

Figure / diagram. Three pillars of the project: Works, Honest, and Open Horizons, accompanied by the open-source software repository card, code metrics, and foundational literature references.



Spoken script. To conclude, I summarize this work across three core pillars.

First: it works. By coupling ZOGY Fourier subtraction with empirical point-spread functions, robust significance scaling, and physical kinematic pairing, SPHERExLens delivers a calibrated detection engine. Strict injection recovery stands at sixty-nine point three percent overall, capturing ninety-seven percent of flaring transients and ninety-five percent of fading sources. Multi-wavelength catalogue triage, asteroid ephemeris checks, and machine learning account for ninety-one percent of all detections.

Second: it is honest. We do not hide behind cherry-picked examples. Split-epoch null tests demonstrate exactly one point six six false candidates per field. Where coverage is inadequate, fields like Sagittarius A* are explicitly refused rather than producing corrupted subtractions. And in our Model Card, we openly disclose that machine-learning headline metrics are inflated by dipoles, reporting true monopole skill transparently.

Third: open horizons. The modular architecture provides a foundation for full-mission deep-field time series, GPU-accelerated spatial deconvolution, and continuous model retraining from public consensus votes.

Everything described today is open-source, fully testable, and cryptographically reproducible. Thank you; I am now delighted to take your questions.

Backup: five technical questions and the pre-flight checklist

Five technical questions and answers.

Q1: Why was the detection threshold calibrated at 6.0σ rather than standard 5.0σ?

Theoretical Gaussian null predictions assume uncorrelated pixels, where 5σ predicts 0.0186 false pixels per frame. However, real co-adds undergo WCS reprojection which correlates noise across ~ 1.5 pixels. In empirical split-epoch null tests across 17 DR1 fields, 5.0σ produced 0.86 false candidates per field, whereas 6.0σ reduced this by 44% to 0.48 per field (and 1.66 overall) with zero penalty to strict injection recovery.

Q2: How does Decision A physical dipole pairing prevent false transient classifications?

In early prototypes, unconstrained $75''$ pairing greedily glued real fading stars to unrelated background troughs, misclassifying them as moving stars. In Decision A, dipole separation is strictly bounded by plausible stellar proper motion over the baseline ($\Delta_{\text{px}} \leq (\mu_{\max}\Delta t)/s$), and the positive and negative lobes must agree within a factor of 2.0 in flux. Fading stars now cleanly become negative monopoles.

Q3: Why are the machine learning model's headline metrics considered inflated by dipoles?

As documented in our Model Card (`ml/MODEL_CARD.md`), synthetic injected movers are dipoles while null-test background artefacts almost never form symmetric dipoles; consequently, the feature `is_dipole` alone separates them. The honest measure of model skill is on monopoles: for brightening sources, precision is 0.90 and recall 0.82; for fading sources, precision is 0.72 and recall 0.63.

Q4: How does the architecture scale from 18 fields to the entire 1.8-million-field sky?

Computationally, ZOGY takes < 0.6 s per 180^2 field, which is trivial on distributed cloud workers. The bottleneck is cutout downloading from IRSA. Our production pipeline solves this by querying the public S3 bucket (`nasa-irsa-spherex`) directly in bulk, hashing catalog listings to skip unchanged sky, and chaining 14-day immutable snapshots in Google Cloud Storage.

Q5: What were the key security and resilience findings resolved in Phase 9?

Our 21-point security review (S1–S21) resolved all identified risks: locking Dockerfiles to unprivileged user `uid 10001`, pinning dependencies with SHA-256 hashed lock files, implementing Netlify S8 HMAC proxy signing (`X-Nf-Sign` HS256 JWT) to prevent header spoofing, bounding background job queues, and sanitising public error responses.

Pre-flight checklist, five minutes before stage presentation.

- **Two browser tabs ready**: (1) Deployed Netlify PWA console at 100% zoom, Barnard's Star loaded; (2) Local fallback on `localhost:8000` backed by cached snapshots.
- **Offline Service Worker primed**: Verify Service Worker status in dev tools; cached history frames ensure smooth presentation if venue Wi-Fi drops.
- **Screen recording fallback**: A clean 1080p MP4 recording of the six demonstration moves saved on desktop in case of live browser hang.
- **This specification document open at Slide 29 (page 30)**: The annotated UI wireframe directly aligns with the browser DOM elements.
- **Hero numbers verified**: 69.3% strict recovery (96.7% brightening, 95.3% fading, 56.0% movers), 1.66 null false candidates/field, 109 detections across 18 DR1 fields (99 KNOWN, 10 CANDIDATE).